# Mathematical Language

Notation and Conventions

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## About This Document

Mathematical Language records the notation and writing conventions adopted here. It is guided by deliberate personal preferences together with established mathematical practice; when no personal convention has been fixed, standard mathematical usage is the default.

This document is intentionally non-exhaustive and may be extended as new subjects and notation are encountered. When multiple alternative notations for the same object or purpose are listed, they are given in preferred order unless stated otherwise: the primary notation is used by default, while listed secondary notation remains acceptable when the surrounding context, an adopted textbook or lecture note, an established field convention, or the notation of a given problem makes it preferable. The absence of a notation from this document does not by itself make that notation invalid.

When notation not covered here is needed, an established standard should be used when suitable; otherwise, a clear, mathematically coherent, and unambiguous notation appropriate to the context may be introduced. Useful choices may later be incorporated into this document.

## Part I. General Conventions and Writing

### 1 General Conventions

Parentheses are preferred for ordinary grouping when delimiters are otherwise interchangeable, including in nested expressions; nesting alone is not a reason to alternate mechanically among parentheses, square brackets, and braces. Square brackets, braces, angle brackets, and other delimiters are used whenever they have an established field-specific, structural, or semantic role.

In LaTeX mathematical source, a punctuation comma is followed by one ordinary source-space, including in tuples, intervals, sets, function arguments, subscripts, and lists of symbols. TeX determines the typeset mathematical spacing; control sequences such as `\,` and syntactic commas in package options are not punctuation commas. Established explicit spacing commands are retained when they serve a typographic purpose.

Notation is selected with regard to established mathematical usage and adopted personal preferences. Distinct concepts may share the same standard symbol, whose meaning is determined by context; such overlap does not by itself require changing either notation. When comparably established choices are available, avoiding ambiguity may be considered without sacrificing mathematical standardness or clarity.

| Item | Convention |
|---|---|
| Scalars | Ordinary mathematical italic letters, for example $a, b, c, \lambda, t$. |
| Vectors and Matrices | Vectors and matrices are written in bold, typically using lowercase and uppercase letters, respectively, such as $\mathbf{u}, \mathbf{v}, \mathbf{x}$ and $\mathbf{A}, \mathbf{B}, \mathbf{M}$. Greek symbols are likewise bold when representing vectors or matrices, such as $\boldsymbol{\xi}, \boldsymbol{\Phi}$. Established field-specific conventions are respected. Unless otherwise stated, vectors are column vectors. |
| Matrix Delimiters | Parentheses are preferred; square brackets are a standard alternative. |
| Dimension subscripts | Prefer dimension subscripts when the dimension is readily available and useful; omit them when it is clear, immaterial, or not readily available rather than inferring a dimension solely to supply a subscript. |
| Number systems | Blackboard bold, for example $\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$. |
| Fields | Arbitrary fields are typically denoted by italic uppercase letters such as $F, K, L$. Established blackboard-bold notation, such as $\mathbb{F}$ and $\mathbb{F}_q$, is also used where appropriate. |
| Sets | Ordinary uppercase letters such as $A, B, X, Y$, unless a field-specific convention requires otherwise. |

| Item | Convention |
|---|---|
| Families and collections | Calligraphic letters when useful, for example $\mathcal{F}, \mathcal{T}, \mathcal{B}$. Script letters such as $\mathscr{A}$ may denote separately defined families or other objects where context warrants; calligraphic and script letters are not interchangeable by default. |
| Symbol distinctions | Different mathematical letterforms and alphabets are treated as distinct symbols, even when based on the same letter. For example, $\varepsilon$ and $\epsilon$, $\varphi$ and $\phi$, and $A$, $\mathrm{A}$, $\mathbf{A}$, $\boldsymbol{A}$, $\mathcal{A}$, $\mathscr{A}$, $\mathbb{A}$, $\mathfrak{A}$, $\mathsf{A}$, and $\mathtt{A}$ are distinct symbols. |
| Named operators | Upright Roman type, for example det, ker, im, rk, and tr. |
| Large operators | Use display-style sizing and conventional index placement by default throughout mathematical writing, including proofs, solutions, examples, inline expressions, and notation tables. This applies to sums, products, indexed unions and intersections, integrals, limits (including limit superior and inferior), extrema, and analogous operators. Indices for sums, products, and similar operators are placed above and below; those for limits and extrema are placed below; integral bounds remain to the right. Smaller sizes or side-positioned indices are permitted only when clearly justified by readability or deliberate typographic layout, such as in fractions or tightly nested expressions, and not merely for convenience or routine space saving. |

| Notation | Meaning |
|---|---|
| $(a_1, \ldots, a_n)$ | Ordered tuple |
| $(a_n),\ \{a_n\}$ | Sequence |
| $(a_i)_{i\in I},\ \{a_i\}_{i\in I}$ | Indexed family over $I$ |
| $(a, b)$ | Open interval |
| $[a, b]$ | Closed interval |
| $[a, b)$ | Left-closed, right-open interval |
| $(a, b]$ | Left-open, right-closed interval |
| $\{a_1, \ldots, a_n\}$ | Set delimiters |
| $\{x \in A \mid P(x)\}$, $\{x \in A\colon P(x)\}$ | Set-builder notation |
| $\langle x, y\rangle$ | Pairing or inner product, according to context |
| $\lvert x\rvert$ | Absolute value, modulus, cardinality, or order, according to context |
| $\lVert x\rVert$ | Norm |
| $a = b$ | Equality |
| $a \coloneqq b$ | $a$ is defined to be $b$. |
| $a \equiv b$ | Identity, equivalence, or congruence, according to context |
| $a \cong b$ | Isomorphism or congruence, according to context |
| $a \approx b$ | Approximation |
| $f\colon X \to Y$ | Function or map from $X$ to $Y$ |
| $x \mapsto f(x)$ | Elementwise rule of a map |
| $f\colon X \to Y,\ x \mapsto f(x)$ | Function with domain, codomain, and mapping rule |
| $g \circ f$ | Composition: first $f$, then $g$ |
| $f\lvert_A,\ f \restriction_A$ | Restriction of $f$ to $A$ |
| $\mathrm{id}_X$ | Identity map on $X$ |

## 2 Mathematical Writing

### Mathematical Environments.

| Item | Convention |
|---|---|
| Definition | Precise meaning of a term, object, property, or construction. |
| Notation | Introduction or declaration of notation. |
| Convention | Standing choice of notation, terminology, normalization, or usage adopted for the stated context. |
| Motivation | Explanation of why a concept, method, problem, or development is introduced or worth considering. |
| Axiom | Fundamental statement accepted without proof as part of a mathematical system or theory. |
| Assumption | Condition adopted as a premise for the current statement, argument, construction, or discussion. |
| Hypothesis | Distinguished condition under which a result or theory is considered; statements conventionally named hypotheses retain that terminology. |
| Conjecture | Statement proposed to be true but not proved in full generality. |
| Theorem | Principal mathematical result. |
| Proposition | Useful result of more limited scope. |
| Lemma | Supporting result, usually for a later proof. |
| Corollary | Result following readily from a preceding result. |
| Claim | Local or intermediate assertion within a larger argument. |
| Fact | Established statement used without the emphasis or status of a named theorem. |
| Observation | Mathematical fact or consequence readily noticed or inferred from the immediate context, definitions, or preceding results. |
| Intuition | Informal or heuristic insight into a mathematical concept, statement, or problem, suggesting what to expect or why a result seems plausible without constituting a proof. |
| Proof | Rigorous argument establishing a stated mathematical assertion. |
| Sketch | Outline of the essential reasoning, construction, or solution with some details omitted. |
| Formula | Distinguished computational identity or closed form. |
| Algorithm | Explicit step-by-step procedure. |
| Construction | Explicit specification of how a mathematical object is defined or built to satisfy stated requirements. |
| Design | Explanation of the roles of individual components and the rationale for structural choices in a mathematical object, function, method, or environment. |
| Example | Illustration of a definition, construction, phenomenon, or result. |
| Counterexample | Example disproving a general assertion or showing that a condition is necessary. |
| Application | Use of a concept, method, or result in a concrete mathematical or applied setting. |
| Remark | Supplementary mathematical clarification, qualification, or extension of the surrounding material. |
| Note | Brief contextual or practical information, such as a local reminder, permission, or usage instruction, worth keeping in mind. |
| Comment | Retrospective expository interpretation or takeaway concerning the significance, lesson, or point of the preceding material. |
| Warning | Important notice of a subtlety, common error, or potentially misleading point. |

| Item | Convention |
|---|---|
| Question | Question posed to prompt thought, exploration, or discussion without necessarily requiring a complete solution. |
| Problem | Mathematical task intended to be solved. |
| Exercise | Task primarily intended to practice, apply, or reinforce mathematical material. |
| Open Problem | Problem not known to be solved in the stated form. |
| Hint | A partial clue intended to help the reader discover a proof or solution without providing the complete argument or answer. |
| Suggestion | An optional recommendation to consider a particular observation, method, or approach when solving a problem or exercise. |
| Solution | Resolution of a stated problem or task by proof, computation, construction, or other appropriate means. |
| Idea | Key insight guiding a proof, solution, or construction. |
| Strategy | Broad problem-solving plan. |
| Tactic | Local step within a larger strategy. |

Proof, sketch, and solution labels are written as *Proof*, *Sketch*, and *Solution*, respectively, each followed by a period.

| Notation | Meaning |
|---|---|
| □ | Ending for ordinary proofs and solutions |
| ■ | Ending for substantial, difficult, or particularly important proofs |
| Q.E.D. | Ending for exceptionally difficult, significant, or paper-level proofs |

**Enumeration and Subdivision.** Enumeration labels are set in upright Roman type. Their form is determined by structural role rather than nesting depth.

| Item | Convention |
|---|---|
| 1.1, 1.2, 1.3, ... | Independent child problems or standalone units under a common parent; each is a distinct subproblem that may be stated, solved, and referenced separately. |
| (1), (2), (3), ... | Ordinary parallel enumeration, such as choices, cases, possibilities, alternatives, examples, or procedural steps; not a subdivision of a single problem or a clause of a formal mathematical statement. |
| (a), (b), (c), ... | Connected subparts of a single problem or task that share a common statement, setup, or context. |
| (i), (ii), (iii), ... | Formal mathematical clauses within definitions, theorems, propositions, lemmas, and related statements, such as hypotheses, conditions, properties, conclusions, or equivalent statements. |

The distinction between 1.1 and (a) is independence: 1.1 denotes a distinct child problem under a common parent, whereas (a) denotes a connected part of one problem sharing its setup. The distinction between (1) and (i) is function: (1) is used for ordinary parallel enumeration, whereas (i) is reserved for clauses within formal mathematical statements. Nested enumeration preserves these structural roles rather than following a fixed hierarchy of label styles.

**Mathematical Shorthand.**

Typeset Abbreviations may be used in mathematical documents, textbooks, problem sets, and solutions, as well as in handwritten work, when appropriate to the context and level of formality. Handwritten Shorthand is reserved for personal notes, scratch work, and blackboard writing.

*Typeset Abbreviations.*

| Notation | Meaning |
|---|---|
| i.e. | that is |
| e.g. | for example |
| etc. | and so on |
| cf. | compare |
| resp. | respectively |
| vs. | versus |
| i.o.w. | in other words |
| s.t. | such that |
| iff | if and only if |
| w.r.t. | with respect to |
| const | constant |
| a.e. | almost everywhere |
| a.s. | almost surely |
| i.i.d. | independent and identically distributed |
| N.t. | Note that |
| w.l.o.g. | without loss of generality |
| w.t.s. | want to show (that) |
| e.t.s. | enough to show (that) |
| t.f.a.e. | the following are equivalent |
| i.t.s.w. | in the same way |
| o.t.o.h. | on the other hand |
| LHS | Left-hand side |
| RHS | Right-hand side |
| IVT | Intermediate Value Theorem |
| EVT | Extreme Value Theorem |
| MVT | Mean Value Theorem |
| FTC | Fundamental Theorem of Calculus |
| ODE | Ordinary Differential Equation |
| PDE | Partial Differential Equation |
| IVP | Initial Value Problem |
| BVP | Boundary Value Problem |

*Handwritten Shorthand.*

| Notation | Meaning |
|---|---|
| Def | Definition |
| Thm | Theorem |
| Prop | Proposition |
| Lem | Lemma |
| Cor | Corollary |
| Pf | Proof |
| Sol | Solution |
| Ex | Example |
| Exer | Exercise |
| Rmk | Remark |
| Sps | Suppose (that) |

| Notation | Meaning |
|---|---|
| cont | continuous |
| cont.ly | continuously |
| difb | differentiable |
| difb.ly | differentiably |
| difl | differential |
| bdd | bounded |
| unif | uniform |
| unif.ly | uniformly |
| lin | linear |
| lin.ly | linearly |
| dep | dependent |
| indep | independent |
| approx | approximate |
| triv | trivial |
| fnc | function |
| seq | sequence |
| eqn | equation |
| eqlt | equality |
| ineqlt | inequality |
| sys | system |
| comb | combination |
| vec | vector |
| mat | matrix |
| col | column |
| pt | point |
| bdy | boundary |
| nbhd | neighborhood |
| itvl | interval |
| op | operation |
| num | number |
| val | value |
| elmt | element |
| cw | clockwise |
| ccw | counterclockwise |
| w/ | with |
| w/o | without |

**Mathematical Phrases.**

| Phrase | Use |
|---|---|
| We find by inspection that ... | Used when a conclusion follows directly from inspection. |

## Part II. Notation by Subject

## 3 Set Theory and Logic

| Notation | Meaning |
|---|---|
| $\mathbb{N} = \{1, 2, 3, \dots\},\ \mathbb{Z}_{\geq 1},\ \mathbb{Z}^+$ | Positive integers |
| $\mathbb{N}_0 = \{0, 1, 2, 3, \dots\},\ \mathbb{Z}_{\geq 0}$ | Nonnegative integers |
| $\mathbb{Z}$ | Integers |
| $\mathbb{Q}$ | Rational numbers |
| $\mathbb{R}$ | Real numbers |
| $\mathbb{C}$ | Complex numbers |
| $\varnothing,\ \emptyset$ | Empty set |
| $x \in A$ | $x$ is an element of $A$. |
| $x \notin A$ | $x$ is not an element of $A$. |
| $A \subseteq B$ | $A$ is a subset of $B$; equality is allowed. |
| $A \subsetneq B$ | $A$ is a proper subset of $B$. |
| $A \supseteq B$ | $A$ is a superset of $B$; equality is allowed. |
| $A \supsetneq B$ | $A$ is a proper superset of $B$. |
| $A \cup B$ | Union |
| $A \cap B$ | Intersection |
| $(A_i)_{i \in I}$ | Indexed family of sets |
| $\bigcup_{i \in I} A_i$ | Union of an indexed family of sets |
| $\bigcap_{i \in I} A_i$ | Intersection of an indexed family of sets |
| $\limsup_{n \to \infty} A_n$ | Set-theoretic limit superior of $(A_n)$: $\bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} A_k$; elements belong to infinitely many $A_n$ |
| $\liminf_{n \to \infty} A_n$ | Set-theoretic limit inferior of $(A_n)$: $\bigcup_{n=1}^{\infty} \bigcap_{k=n}^{\infty} A_k$; elements belong to all but finitely many $A_n$ |
| $A \setminus B,\ A - B$ | Set difference |
| $A \triangle B,\ A\, \Delta\, B$ | Symmetric difference |
| $A^c$ | Complement of $A$ in the ambient set |
| $\mathcal{P}(A),\ 2^A$ | Power set |
| $A \times B$ | Cartesian product |
| $A_1 \times \cdots \times A_n,\ A^n$ | Finite Cartesian product; $A^n$ is the $n$-fold Cartesian power of $A$ |
| $\prod_{i \in I} A_i$ | Cartesian product of an indexed family of sets |
| $A \sqcup B,\ A \dot\cup B$ | Disjoint union |
| $\bigsqcup_{i \in I} A_i$ | Disjoint union of an indexed family of sets |
| $A \cap B = \varnothing$ | $A$ and $B$ are disjoint. |
| $R \subseteq X \times Y$ | Binary relation between $X$ and $Y$ |
| $xRy$ | $x$ is related to $y$ by the relation $R$. |
| $R^{-1}$ | Inverse relation of $R$ |
| $[x]_R,\ [x]$ | Equivalence class of $x$ under $R$ |
| $x \sim y$ | Equivalent elements under an understood equivalence relation |
| $A/\!\sim$ | Set of equivalence classes |
| $\operatorname{dom} f$ | Domain of $f$ |
| $\operatorname{codom} f$ | Codomain of $f$ |

| Notation | Meaning |
|---|---|
| $f(A)$ | Image of $A$ under $f$ |
| $f^{-1}(B)$ | Preimage of $B$ under $f$; this notation does not require $f$ to be invertible |
| $\operatorname{im} f$ | Image of the map $f$ |
| $Y^X$ | Set of all functions from $X$ to $Y$ |
| $f^{-1}\colon Y \to X$ | Inverse function of a bijection $f\colon X \to Y$ |
| $f\colon X \hookrightarrow Y$ | Injective map |
| $f\colon X \twoheadrightarrow Y$ | Surjective map |
| $f\colon X \rightarrowtail Y$ | Bijective map |
| $f\colon X \xrightarrow{\sim} Y$ | Isomorphism; for sets, equivalently a bijection |
| $\mathbb{1}_A(x)$ | Indicator function of $A$ |
| $\forall x \in A$ | For every $x \in A$ |
| $\exists x \in A$ | There exists $x \in A$. |
| $\exists! x \in A$ | There exists a unique $x \in A$. |
| $\nexists x \in A$ | There does not exist $x \in A$. |
| $P \Rightarrow Q$ | $P$ implies $Q$ in ordinary mathematical writing. |
| $P \Leftarrow Q$ | $Q$ implies $P$ in ordinary mathematical writing. |
| $P \not\Rightarrow Q$ | $P$ does not imply $Q$. |
| $P \Leftrightarrow Q$ | $P$ if and only if $Q$ in ordinary mathematical writing. |
| $P \wedge Q$ | Logical conjunction |
| $P \vee Q$ | Logical disjunction |
| $\neg P$ | Logical negation |
| $\top,\ \bot$ | Truth and falsehood constants, respectively, in formal logic |
| $\varphi \to \psi,\ \varphi \leftrightarrow \psi$ | Conditional and biconditional connectives in formal logic |
| $\mathcal{M} \models \varphi,\ \mathcal{M} \not\models \varphi$ | $\mathcal{M}$ satisfies, or does not satisfy, the formula $\varphi$. |
| $\Gamma \models \varphi,\ \Gamma \not\models \varphi$ | $\varphi$ is, or is not, a semantic consequence of $\Gamma$. |
| $\Gamma \vdash \varphi,\ \Gamma \nvdash \varphi$ | $\varphi$ is, or is not, syntactically derivable from $\Gamma$. |
| $\lightning$ | Contradiction reached in a proof or argument; distinct from the formal falsehood constant $\bot$ |
| $\therefore$ | Therefore |
| $\because$ | Because |
| $\lvert A\rvert,\ \#A$ | Cardinality |
| $\aleph_0$ | Cardinality of a countably infinite set |
| $\mathfrak{c}$ | Cardinality of the continuum |

## 4 Calculus and Analysis

The differential $\mathrm{d}$ is upright; $\varepsilon$ is the default small positive quantity in analysis.

| Notation | Meaning |
|---|---|
| $\lim_{x \to a} f(x),\ f(a^{\pm})$ | Two-sided limit at $a$; $f(a^{\pm})$ denotes the common one-sided limit (adopted convention) |
| $\lim_{x \to a^-} f(x),\ f(a^-)$ | Left-hand limit at $a$ |
| $\lim_{x \to a^+} f(x),\ f(a^+)$ | Right-hand limit at $a$ |
| $\lim_{x \to \infty} f(x),\ f(\infty)$ | Limit as $x \to \infty$; $f(\infty)$ is adopted shorthand when the limit exists |
| $\lim_{x \to -\infty} f(x),\ f(-\infty)$ | Limit as $x \to -\infty$; $f(-\infty)$ is adopted shorthand when the limit exists |

| Notation | Meaning |
|---|---|
| $\lim_{x\to a} f(x) = \infty$, $\lim_{x\to a} f(x) = -\infty$ | Infinite limits |
| $\min A$ | Minimum of $A$ |
| $\max A$ | Maximum of $A$ |
| $\inf A$ | Infimum of $A$ |
| $\sup A$ | Supremum of $A$ |
| $\arg\min_{x\in A} f(x)$ | Set of minimizers of $f$ over $A$; empty if the infimum is not attained |
| $\arg\max_{x\in A} f(x)$ | Set of maximizers of $f$ over $A$; empty if the supremum is not attained |
| $\overline{\mathbb{R}} = \mathbb{R} \cup \{-\infty, +\infty\}$ | Extended real line |
| $f'(a)$ | First derivative of $f$ at $a$ |
| $f^{(n)}(a)$ | $n$th derivative of $f$ at $a$ |
| $\dfrac{\mathrm{d}y}{\mathrm{d}x}$ | Derivative of $y$ with respect to $x$ |
| $\dfrac{\mathrm{d}^n y}{\mathrm{d}x^n}$ | $n$th derivative of $y$ with respect to $x$ in Leibniz notation |
| $\dfrac{\partial f}{\partial x_i},\ \partial_i f,\ f_{x_i}$ | Partial derivative with respect to $x_i$ |
| $\dfrac{\partial^2 f}{\partial x_j \partial x_i},\ \partial_{ji} f,\ f_{x_i x_j}$ | Successive or mixed partial derivative, first with respect to $x_i$ and then $x_j$. When the relevant mixed partial derivatives are continuous, the order may be interchanged. |
| $D_{\mathbf{v}} f(\mathbf{a}),\ \partial_{\mathbf{v}} f(\mathbf{a})$ | Directional derivative along the vector $\mathbf{v}$, which need not be a unit vector: $D_{\mathbf{v}} f(\mathbf{a}) = \dfrac{\mathrm{d}}{\mathrm{d}t} f(\mathbf{a} + t\mathbf{v})\Big\vert_{t=0} = Df(\mathbf{a})[\mathbf{v}] = \nabla f(\mathbf{a}) \cdot \mathbf{v}$ when $f$ is differentiable. For the unit direction determined by $\mathbf{v} \neq \mathbf{0}$, use $\mathbf{u} = \mathbf{v}/\lVert\mathbf{v}\rVert$ |
| $\mathrm{d}f = \sum_{i=1}^{n} \dfrac{\partial f}{\partial x_i}\ \mathrm{d}x_i$ | Total differential of a scalar function of $n$ variables |
| $\int f(x)\ \mathrm{d}x$ | Indefinite integral; a family of antiderivatives, with an additive constant understood |
| $\int_a^b f(x)\ \mathrm{d}x$ | Definite integral over $[a, b]$ |
| $\int_a^\infty f(x)\ \mathrm{d}x,\ \int_{-\infty}^b f(x)\ \mathrm{d}x$ | Improper integrals over unbounded intervals |
| $\int_a^b f\ \mathrm{d}\alpha$ | Riemann–Stieltjes integral of $f$ with respect to the integrator $\alpha$; in the classical setup here, $\alpha\colon [a, b] \to \mathbb{R}$ is nondecreasing, not necessarily continuous |
| $P = \{x_0, \ldots, x_n\}$ | Partition of $[a, b]$, with $a = x_0 < \cdots < x_n = b$ |
| $\sum_{i=1}^{n} f(\xi_i)\Delta x_i$ | Tagged Riemann sum, where $\Delta x_i = x_i - x_{i-1}$ and $\xi_i \in [x_{i-1}, x_i]$ |
| $\iint_D f\ \mathrm{d}A$ | Double integral over $D$ |
| $\iiint_\Omega f\ \mathrm{d}V$ | Triple integral over $\Omega$ |
| $\int_C f(z)\ \mathrm{d}z$ | Contour integral along an oriented contour $C$ |
| $\oint_C f(z)\ \mathrm{d}z$ | Contour integral over a closed contour $C$ |
| $\mathrm{Ind}(\gamma, z)$ | Winding number (index) of a closed piecewise $C^1$ curve $\gamma$ about $z \notin \gamma([a, b])$: $\dfrac{1}{2\pi i} \displaystyle\int_\gamma \frac{\mathrm{d}w}{w - z}$ |

| Notation | Meaning |
|---|---|
| $-C$ | The contour $C$ with the opposite orientation |
| $\gamma^{-1}$ | Reversed parametrized path $\gamma$, when unambiguous; not a functional inverse |
| PV, P.V. | Principal value; in particular, the Cauchy principal value of a singular or improper integral |
| $\sum_{k=1}^{n} a_k$ | Finite sum |
| $\sum_{k=1}^{\infty} a_k$ | Infinite series |
| $\prod_{k=1}^{n} a_k$ | Finite product |
| $\sum_{n=0}^{\infty} a_n(x-a)^n$ | Power series centered at $a$ |
| $\sum_{n=-\infty}^{\infty} a_n(z-z_0)^n$ | Laurent series about $z_0$, on an annulus of convergence |
| $\sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k$ | $n$th Taylor polynomial of $f$ about $a$ |
| $\widehat{f}(n)$ | $n$th complex Fourier coefficient of an integrable $2\pi$-periodic function $f$: $\widehat{f}(n) = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)e^{-inx}\,\mathrm{d}x$, for $n \in \mathbb{Z}$ |
| $S_N f$ | $N$th symmetric Fourier partial sum under this convention: $S_N f(x) = \sum_{n=-N}^{N} \widehat{f}(n)e^{inx}$, for $N \geq 0$ |
| $\sigma_N f$ | $N$th Cesàro (Fejér) mean of the Fourier partial sums: $\sigma_N f(x) = \frac{1}{N+1}\sum_{k=0}^{N} S_k f(x)$, for $N \geq 0$ |
| $e^x$, $\exp x$ | Exponential function |
| $\ln x$, $\log_e x$ | Natural logarithm of a positive real number |
| $\lg x$, $\log_{10} x$ | Logarithm to base 10 |
| $\mathrm{lb}\, x$, $\log_2 x$ | Logarithm to base 2 |
| $\log_a x$ | Logarithm to base $a$ |
| $W_k(z)$ | Lambert $W$-function: $W_k(z)e^{W_k(z)} = z$; $W_0$ is the principal branch |
| $\sin x$ | Sine |
| $\cos x$ | Cosine |
| $\tan x$ | Tangent |
| $\csc x$ | Cosecant |
| $\sec x$ | Secant |
| $\cot x$ | Cotangent |
| $\arcsin x$, $\sin^{-1} x$ | Principal inverse sine, $[-1,1] \to [-\pi/2, \pi/2]$ |
| $\arccos x$, $\cos^{-1} x$ | Principal inverse cosine, $[-1,1] \to [0,\pi]$ |
| $\arctan x$, $\tan^{-1} x$ | Principal inverse tangent, $\mathbb{R} \to (-\pi/2, \pi/2)$ |
| $\operatorname{arccsc} x$, $\csc^{-1} x$ | Principal inverse cosecant, defined by $\operatorname{arccsc} x = \arcsin(1/x)$ for $\lvert x\rvert \geq 1$; its range is $[-\pi/2, 0) \cup (0, \pi/2]$ |
| $\operatorname{arcsec} x$, $\sec^{-1} x$ | Principal inverse secant, defined by $\operatorname{arcsec} x = \arccos(1/x)$ for $\lvert x\rvert \geq 1$; its range is $[0, \pi/2) \cup (\pi/2, \pi]$ |
| $\operatorname{arccot} x$, $\cot^{-1} x$ | Principal real inverse cotangent, $\mathbb{R} \to (0, \pi)$ |
| $\sinh x$ | Hyperbolic sine |

| Notation | Meaning |
|---|---|
| $\cosh x$ | Hyperbolic cosine |
| $\tanh x$ | Hyperbolic tangent |
| $\operatorname{csch} x$ | Hyperbolic cosecant |
| $\operatorname{sech} x$ | Hyperbolic secant |
| $\coth x$ | Hyperbolic cotangent |
| $\operatorname{arcsinh} x,\ \sinh^{-1} x$ | Principal real inverse hyperbolic sine, $\mathbb{R} \to \mathbb{R}$ |
| $\operatorname{arccosh} x,\ \cosh^{-1} x$ | Principal real inverse hyperbolic cosine, $[1, \infty) \to [0, \infty)$ |
| $\operatorname{arctanh} x,\ \tanh^{-1} x$ | Principal real inverse hyperbolic tangent, $(-1, 1) \to \mathbb{R}$ |
| $\operatorname{arccsch} x,\ \operatorname{csch}^{-1} x$ | Principal real inverse hyperbolic cosecant, defined by $\operatorname{arccsch} x = \operatorname{arcsinh}(1/x)$ on $\mathbb{R} \setminus \{0\}$; its range is $\mathbb{R} \setminus \{0\}$ |
| $\operatorname{arcsech} x,\ \operatorname{sech}^{-1} x$ | Principal real inverse hyperbolic secant, defined by $\operatorname{arcsech} x = \operatorname{arccosh}(1/x)$ on $(0, 1]$; its range is $[0, \infty)$ |
| $\operatorname{arccoth} x,\ \coth^{-1} x$ | Principal real inverse hyperbolic cotangent, defined by $\operatorname{arccoth} x = \operatorname{arctanh}(1/x)$ for $\lvert x \rvert > 1$; its range is $\mathbb{R} \setminus \{0\}$ |
| $\Gamma(z)$ | Gamma function |
| $\psi(z)$ | Digamma function, $\psi(z) = \Gamma'(z)/\Gamma(z)$ |
| $B(x, y)$ | Beta function |
| $\gamma$ | Euler–Mascheroni constant, $\gamma = \lim_{n\to\infty} \left( \sum_{k=1}^{n} \frac{1}{k} - \ln n \right)$ |
| $\zeta(s)$ | Riemann zeta function |
| $\mathbf{J_F}(\mathbf{a}),\ D\mathbf{F}(\mathbf{a})$ | Jacobian matrix of $\mathbf{F} = (F_1, \ldots, F_m) \colon \mathbb{R}^n \to \mathbb{R}^m$ at $\mathbf{a}$, with $\mathbf{J_F}(\mathbf{a}) = \left( \frac{\partial F_i}{\partial x_j}(\mathbf{a}) \right)_{i,j}$. $D\mathbf{F}(\mathbf{a})$ is the total derivative in standard coordinates |
| $\det \mathbf{J_F}(\mathbf{a}),\ \det D\mathbf{F}(\mathbf{a})$ | Jacobian determinant of $\mathbf{F} \colon \mathbb{R}^n \to \mathbb{R}^n$ at $\mathbf{a}$; also written $\left. \frac{\partial(F_1, \ldots, F_n)}{\partial(x_1, \ldots, x_n)} \right\rvert_{\mathbf{a}}$ |
| $\operatorname{Hess} f$ | Hessian matrix of $f$ |
| $\nabla f,\ \operatorname{grad} f$ | Gradient of $f$ |
| $\nabla \cdot \mathbf{F},\ \operatorname{div} \mathbf{F}$ | Divergence of $\mathbf{F}$ |
| $\nabla \times \mathbf{F},\ \operatorname{curl} \mathbf{F}$ | Curl of $\mathbf{F}$ |
| $\Delta f$ | Laplacian of $f$ |
| $\mathbf{r}(t)$ | Parametrized curve |
| $\mathrm{d}s$ | Element of arc length |
| $\int_C f \, \mathrm{d}s$ | Scalar line integral along $C$ with respect to arc length |
| $\int_C \mathbf{F} \cdot \mathrm{d}\mathbf{r}$ | Line integral of the vector field $\mathbf{F}$ along $C$ |
| $\iint_S f \, \mathrm{d}S$ | Scalar surface integral over $S$ |
| $\iint_S \mathbf{F} \cdot \mathbf{n} \, \mathrm{d}S$ | Flux integral through $S$ |
| $x_n \to x,\ \lim_{n\to\infty} x_n = x$ | Convergence of a sequence |
| $(x_{n_k})$ | Subsequence of the sequence $(x_n)$ |
| $a_n \nearrow L$ | Monotone increasing convergence to $L$ |
| $a_n \searrow L$ | Monotone decreasing convergence to $L$ |
| $\limsup_{n\to\infty} a_n$ | Limit superior of the sequence $(a_n)$ |
| $\liminf_{n\to\infty} a_n$ | Limit inferior of the sequence $(a_n)$ |

| Notation | Meaning |
|---|---|
| $f(x) \sim g(x)$ | Asymptotic equivalence in the stated limiting regime, typically meaning $f(x)/g(x) \to 1$ |
| $f_n \to f,\ f_n \xrightarrow{\text{ptw}} f$ | Pointwise convergence of functions on a common domain by default when no ambient topology, metric, or norm is specified. If such a structure is specified, the ordinary arrow $\to$ refers to convergence in that structure |
| $f_n \rightrightarrows f,\ f_n \xrightarrow{\text{unif}} f$ | Uniform convergence |
| $f_n \xrightarrow{\text{a.e.}} f$ | Convergence almost everywhere |
| $(X, \mathcal{A}, \mu)$ | Measure space, where $\mathcal{A}$ is a $\sigma$-algebra on $X$ and $\mu$ is a measure on $\mathcal{A}$ |
| $\mathcal{A}$ | $\sigma$-algebra of measurable subsets of $X$, when the underlying set is understood |
| $\mathcal{B}(X)$ | Borel $\sigma$-algebra of a topological space $X$ |
| $\mu(E)$ | Measure of a measurable set $E \in \mathcal{A}$ |
| $\displaystyle\int_E f \, \mathrm{d}\mu$ | Integral of $f$ over $E$ with respect to the measure $\mu$ |
| $f = g$ a.e., $\quad f = g$ $\mu$-a.e. | Equality almost everywhere, with the measure omitted when understood |
| $\operatorname{ess\,sup}_{x \in E} f(x)$ | Essential supremum of $f$ on $E$ |
| $\operatorname{ess\,inf}_{x \in E} f(x)$ | Essential infimum of $f$ on $E$ |
| $C(\Omega),\ C^0(\Omega)$ | Continuous functions on $\Omega$ |
| $C^k(\Omega)$ | $k$ times continuously differentiable functions on $\Omega$ |
| $C^\infty(\Omega)$ | Smooth functions on $\Omega$ |
| $C_b(X)$ | Bounded continuous scalar-valued functions on $X$; the scalar field is given by context, or explicitly by $C_b(X, \mathbb{R})$ or $C_b(X, \mathbb{C})$. If $K$ is compact, $C_b(K) = C(K)$ for the same codomain |
| $\operatorname{supp} f$ | Support of $f$, the closure of the set on which $f$ is nonzero |
| $C_c(\Omega),\ C_c^\infty(\Omega)$ | Continuous and smooth functions on $\Omega$ with compact support, respectively |
| $\mathcal{O}(U)$ | Holomorphic functions on an open set $U \subseteq \mathbb{C}$ |
| $V_a^b(f)$ | Total variation of $f$ on $[a, b]$: $\sup_P \sum_{i=1}^{n} \lvert f(x_i) - f(x_{i-1}) \rvert$, over partitions $P = (a = x_0 < \cdots < x_n = b)$ |
| $\mathrm{BV}([a, b])$ | Space of functions of bounded variation on $[a, b]$, namely those $f$ with $V_a^b(f) < \infty$ |
| $\mathcal{R}([a, b])$ | Riemann-integrable scalar-valued functions on $[a, b]$; equivalently $\mathcal{R}([a, b], \mathrm{id})$ for the identity integrator $\mathrm{id}(x) = x$ |
| $\mathcal{R}([a, b], \alpha)$ | Riemann–Stieltjes-integrable scalar-valued functions on $[a, b]$ with respect to a nondecreasing real-valued function $\alpha\colon [a, b] \to \mathbb{R}$; $\alpha$ need not be continuous, and complex-valued integrands are handled componentwise |
| $L^p(\Omega),\ L^p(X, \mu)$ | Lebesgue $L^p$ spaces of measurable functions modulo equality almost everywhere; the measure is displayed when not understood. For $1 \le p < \infty$, membership requires $\displaystyle\int_X \lvert f \rvert^p \, \mathrm{d}\mu < \infty$ |
| $L^\infty(\Omega),\ L^\infty(X, \mu)$ | Lebesgue spaces of essentially bounded measurable functions, identified up to equality almost everywhere |
| $\ell^p$ | Space of scalar sequences $(x_n)$ with $\displaystyle\sum_{n=1}^{\infty} \lvert x_n \rvert^p < \infty$, for $1 \le p < \infty$ |
| $\ell^\infty$ | Space of bounded scalar sequences |

| Notation | Meaning |
|---|---|
| $\|f\|_\infty$ | Supremum norm $\sup_{x\in X} \|f(x)\|$ for an actual bounded scalar-valued function on $X$ |
| $\|f\|_{L^p(X,\mu)}$ | $L^p$-norm $\left(\int_X \|f\|^p \, \mathrm{d}\mu\right)^{1/p}$ for $1 \le p < \infty$ on a measure space $(X, \mathcal{A}, \mu)$ |
| $\|f\|_{L^p(\Omega)},\ \|f\|_p$ | $L^p$-norm $\left(\int_\Omega \|f(\mathbf{x})\|^p \, \mathrm{d}\mathbf{x}\right)^{1/p}$ for $1 \le p < \infty$ on a Euclidean domain when the underlying measure is understood |
| $\|f\|_{L^\infty(\Omega)}$ | $L^\infty$-norm $\operatorname{ess\,sup}_{\mathbf{x}\in\Omega} \|f(\mathbf{x})\|$ on a Euclidean domain; distinguished from the ordinary supremum norm |
| $\alpha = (\alpha_1, \ldots, \alpha_n) \in \mathbb{N}_0^n$ | Multi-index |
| $\|\alpha\|,\ \alpha!$ | Order $\|\alpha\| = \alpha_1 + \cdots + \alpha_n$ and factorial $\alpha! = \alpha_1! \cdots \alpha_n!$ of a multi-index |
| $\mathbf{x}^\alpha = x_1^{\alpha_1} \cdots x_n^{\alpha_n}$ | Multi-index monomial for $\mathbf{x} = (x_1, \ldots, x_n)$ |
| $D^\alpha f,\ \partial^\alpha f$ | Mixed partial derivative specified by the multi-index $\alpha$ |
| $B_r(x)$ | Open ball of radius $r$ centered at $x$ |
| $\overline{B}_r(x) = \{y \mid d(x,y) \le r\}$ | Closed ball of radius $r$ centered at $x$ in a metric space $(X, d)$. The bar is part of the closed-ball notation; in an arbitrary metric space, $\overline{B}_r(x)$ need not equal the closure $\overline{B_r(x)}$ of the open ball |
| $\widehat{\mathbb{C}} = \mathbb{C} \cup \{\infty\}$ | Extended complex plane, equivalently the Riemann sphere |
| $i^2 = -1$ | Imaginary unit |
| $z = x + iy$ | Complex number with real part $x$ and imaginary part $y$ |
| $\operatorname{Re} z,\ \Re z$ | Real part of $z$ |
| $\operatorname{Im} z,\ \Im z$ | Imaginary part of $z$; distinct from lowercase im for image |
| $\overline{z}$ | Complex conjugate |
| $\|z\|$ | Complex modulus |
| $\arg z$ | Argument of a nonzero complex number $z$; its values differ by integer multiples of $2\pi$ |
| $\operatorname{Arg} z$ | Principal argument of $z \neq 0$, with $-\pi < \operatorname{Arg} z \le \pi$ |
| $\log z,\ \ln z$ | General multivalued complex logarithm for $z \neq 0$; $\log z = \{\ln \|z\| + i(\operatorname{Arg} z + 2\pi k) \mid k \in \mathbb{Z}\}$ |
| $\operatorname{Log} z,\ \operatorname{Ln} z$ | Principal complex logarithm for $z \neq 0$; $\operatorname{Log} z = \ln \|z\| + i \operatorname{Arg} z$. On $\mathbb{C} \setminus (-\infty, 0]$, it is the analytic principal branch, with $-\pi < \operatorname{Arg} z < \pi$ |
| $\operatorname{Res}(f; a),\ \operatorname{Res}_{z=a} f(z)$ | Residue of $f$ at a finite point $a$ |
| $\operatorname{Res}(f; \infty),\ \operatorname{Res}_{z=\infty} f(z)$ | Residue of $f$ at infinity |

## 5 Linear Algebra

| Notation | Meaning |
|---|---|
| $M_{m\times n}(F),\ F^{m\times n}$ | Space of $m \times n$ matrices over the field $F$ |
| $M_n(F)$ | Space (and matrix algebra) of $n \times n$ matrices over $F$ |
| $\mathbf{A} = (a_{ij}),\ \mathbf{A} = [a_{ij}]$ | Matrix with $(i,j)$-entry $a_{ij}$ |
| $\mathbf{A}^\mathsf{T}$ | Transpose of $\mathbf{A}$ |
| $\mathbf{A}^\mathsf{H},\ \mathbf{A}^*$ | Conjugate transpose of $\mathbf{A}$ |
| $\mathbf{A}^{-1}$ | Inverse of $\mathbf{A}$ |
| $\mathbf{I}_n,\ \mathbf{I}$ | Identity matrix |
| $\mathbf{0}$ | Zero vector or zero matrix, with dimension understood |

| Notation | Meaning |
|---|---|
| $\mathbf{e}_i$ | $i$th standard basis vector |
| $\mathbf{1}$ | All-ones vector |
| $\mathbf{D}$ | Diagonal matrix |
| $\operatorname{diag}(d_1, \ldots, d_n)$ | Diagonal matrix with entries $d_1, \ldots, d_n$ |
| $\mathbf{P}$ | Projection matrix |
| $\det(\mathbf{A}),\ \lvert\mathbf{A}\rvert$ | Determinant |
| $\operatorname{rk}(\mathbf{A}),\ \operatorname{rank}(\mathbf{A})$ | Rank |
| $\operatorname{tr}(\mathbf{A})$ | Trace |
| $\operatorname{adj}(\mathbf{A})$ | Adjugate matrix |
| $M_{ij}$ | Minor of the $(i,j)$-entry |
| $C_{ij} = (-1)^{i+j} M_{ij}$ | Cofactor of the $(i,j)$-entry |
| $\operatorname{Col}(\mathbf{A})$ | Column space |
| $\operatorname{Row}(\mathbf{A})$ | Row space |
| $\ker \mathbf{A},\ \operatorname{Null}(\mathbf{A})$ | Null space of $\mathbf{A}$ |
| $V, W$ | Vector spaces, usually over an understood field $F$ |
| $\operatorname{Hom}_F(V, W)$ | Vector space of $F$-linear maps from $V$ to $W$ |
| $\mathcal{L}(X, Y)$ | Bounded linear maps between normed spaces $X$ and $Y$ |
| $\lVert T\rVert_{\text{op}}$ | Operator norm $\sup_{\lVert x\rVert \le 1} \lVert Tx\rVert$ of a bounded linear map $T\colon X \to Y$ |
| $\lVert\mathbf{A}\rVert_{\text{F}}$ | Frobenius norm of $\mathbf{A}$ |
| $\mathbf{A} \otimes \mathbf{B}$ | Kronecker product of matrices $\mathbf{A}$ and $\mathbf{B}$ |
| $\operatorname{End}_F(V)$ | $F$-algebra of $F$-linear endomorphisms of $V$ |
| $\dim V$ | Dimension of $V$ |
| $\operatorname{span}(\mathbf{v}_1, \ldots, \mathbf{v}_k)$ | Span of the listed vectors |
| $U \oplus V$ | Direct sum |
| $V/U$ | Quotient vector space of $V$ by its vector subspace $U \subseteq V$; elements are cosets $\mathbf{v} + U$ |
| $\ker T$ | Kernel of a linear map $T$ |
| $\operatorname{nullity}(T) = \dim \ker T$ | Nullity of a linear map $T$ |
| $\operatorname{im} T$ | Image of a linear map $T$ |
| $\mathcal{B} = (\mathbf{b}_1, \ldots, \mathbf{b}_n)$ | Ordered basis |
| $[\mathbf{v}]_{\mathcal{B}}$ | Coordinate vector of $\mathbf{v}$ with respect to the ordered basis $\mathcal{B}$ |
| $[T]_{\mathcal{C} \leftarrow \mathcal{B}}$ | Matrix of $T\colon V \to W$ with respect to domain basis $\mathcal{B}$ and codomain basis $\mathcal{C}$; thus $[T(\mathbf{v})]_{\mathcal{C}} = [T]_{\mathcal{C} \leftarrow \mathcal{B}} [\mathbf{v}]_{\mathcal{B}}$ |
| $[T]_{\mathcal{B}}$ | Matrix of an endomorphism $T\colon V \to V$ when the same basis $\mathcal{B}$ is used in the domain and codomain |
| $\mathbf{P}_{\mathcal{C} \leftarrow \mathcal{B}}$ | Change-of-coordinate matrix from $\mathcal{B}$-coordinates to $\mathcal{C}$-coordinates; $[\mathbf{v}]_{\mathcal{C}} = \mathbf{P}_{\mathcal{C} \leftarrow \mathcal{B}} [\mathbf{v}]_{\mathcal{B}}$ |
| $V^*$ | Algebraic dual of a vector space $V$; for a normed space $X$, $X^*$ usually denotes the continuous dual |
| $\langle \mathbf{u}, \mathbf{v} \rangle,\ (\mathbf{u}, \mathbf{v})$ | Inner product. Over complex spaces, the inner product is conjugate-linear in the first argument and linear in the second |
| $\mathbf{u} \cdot \mathbf{v},\ \mathbf{u}^{\mathsf{T}}\mathbf{v}$ | Standard Euclidean inner product on $\mathbb{R}^n$ |
| $\mathbf{u}^{\mathsf{H}}\mathbf{v}$ | Standard Hermitian inner product on $\mathbb{C}^n$ |
| $\lVert\mathbf{v}\rVert$ | Norm of $\mathbf{v}$ |
| $\mathbf{u} \perp \mathbf{v}$ | Orthogonality |
| $U^{\perp}$ | Orthogonal complement of the subspace $U$ |
| $\operatorname{proj}_U \mathbf{x}$ | Orthogonal projection of $\mathbf{x}$ onto the subspace $U$ |

| Notation | Meaning |
|---|---|
| $\text{proj}_{\mathbf{v}}\,\mathbf{x}$ | Orthogonal projection of $\mathbf{x}$ onto span($\mathbf{v}$) |
| $\mathbf{u} \times \mathbf{v}$ | Cross product in $\mathbb{R}^3$ |
| $\mathbf{A}^\mathsf{T} = \mathbf{A}$ | Real symmetric matrix |
| $\mathbf{A}^\mathsf{T} = -\mathbf{A}$ | Real skew-symmetric matrix |
| $\mathbf{A}^\mathsf{H} = \mathbf{A}$ | Hermitian matrix |
| $\mathbf{A}^\mathsf{H} = -\mathbf{A}$ | Skew-Hermitian matrix |
| $\mathbf{Q}^\mathsf{T}\mathbf{Q} = \mathbf{I}$ | Real orthogonal matrix $\mathbf{Q}$ |
| $\mathbf{U}^\mathsf{H}\mathbf{U} = \mathbf{I}$ | Unitary matrix $\mathbf{U}$ |
| $\mathbf{A}\mathbf{A}^\mathsf{H} = \mathbf{A}^\mathsf{H}\mathbf{A}$ | Normal matrix |
| $\mathbf{P}^2 = \mathbf{P}$ | Projection matrix |
| $\mathbf{P}^2 = \mathbf{P} = \mathbf{P}^\mathsf{T}$ | Orthogonal projection matrix over $\mathbb{R}$ |
| $\mathbf{P}^2 = \mathbf{P} = \mathbf{P}^\mathsf{H}$ | Orthogonal projection matrix over $\mathbb{C}$ |
| $\chi_\mathbf{A}(t),\ p_\mathbf{A}(t)$ | Characteristic polynomial of $\mathbf{A}$, with $\chi_\mathbf{A}(t) = \det(t\mathbf{I} - \mathbf{A})$ |
| $m_\mathbf{A}(t),\ \mu_\mathbf{A}(t)$ | Minimal polynomial of $\mathbf{A}$ |
| $\lambda,\ \lambda_i$ | Eigenvalue of $\mathbf{A}$ |
| $\mathbf{v},\ \mathbf{v}_i$ | Eigenvector of $\mathbf{A}$ |
| $E_\lambda(\mathbf{A}),\ V_\lambda(\mathbf{A})$ | Eigenspace associated with $\lambda$; $E_\lambda(\mathbf{A}) = \ker(\mathbf{A} - \lambda\mathbf{I})$ |
| $\mathbf{J}_k(\lambda)$ | Jordan block of size $k$ with eigenvalue $\lambda$, having $\lambda$ on the diagonal and ones on the superdiagonal |
| $\sigma(\mathbf{A}),\ \text{spec}(\mathbf{A})$ | Spectrum of $\mathbf{A}$ |
| $\rho(\mathbf{A})$ | Spectral radius |
| $\mathbf{P}_\lambda$ | Spectral projection associated with the eigenvalue $\lambda$, when defined |
| $\mathbf{A} = \mathbf{P}\mathbf{D}\mathbf{P}^{-1}$ | Diagonalization of a diagonalizable matrix |
| $\mathbf{A} = \mathbf{Q}\mathbf{D}\mathbf{Q}^\mathsf{T}$ | Orthogonal diagonalization of a real symmetric matrix, with $\mathbf{Q}^\mathsf{T}\mathbf{Q} = \mathbf{I}$ |
| $\mathbf{A} = \mathbf{U}\mathbf{D}\mathbf{U}^\mathsf{H}$ | Unitary diagonalization of a complex normal matrix, with $\mathbf{U}^\mathsf{H}\mathbf{U} = \mathbf{I}$ |
| $\mathbf{A} = \sum_{\lambda \in \sigma(\mathbf{A})} \lambda \mathbf{P}_\lambda$ | Spectral decomposition of a diagonalizable matrix, with the sum over distinct eigenvalues |
| $e^\mathbf{A},\ \exp(\mathbf{A})$ | Matrix exponential |
| $\mathbf{A} = \mathbf{L}\mathbf{U}$ | LU factorization, with $\mathbf{L}$ lower triangular and $\mathbf{U}$ upper triangular |
| $\mathbf{A} = \mathbf{Q}\mathbf{R}$ | QR factorization |
| $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\mathsf{H}$ | Singular value decomposition (SVD); over $\mathbb{R}$, $\mathbf{V}^\mathsf{H} = \mathbf{V}^\mathsf{T}$ |
| $\sigma_i(\mathbf{A})$ | $i$th singular value of $\mathbf{A}$ |
| $\mathbf{A}^\dagger$ | Moore–Penrose pseudoinverse of $\mathbf{A}$ |
| $\mathbf{A} \succ 0$ | Positive definite Hermitian matrix (real symmetric in the real case) |
| $\mathbf{A} \succeq 0$ | Positive semidefinite Hermitian matrix |
| $\mathbf{A} \prec 0$ | Negative definite Hermitian matrix |
| $\mathbf{A} \preceq 0$ | Negative semidefinite Hermitian matrix |
| $\mathbf{A} \succeq \mathbf{B}$ | Löwner order: $\mathbf{A} - \mathbf{B} \succeq 0$, for Hermitian matrices of the same size |
| $\mathbf{A} = \mathbf{L}\mathbf{L}^\mathsf{H}$ | Cholesky factorization of a positive definite Hermitian matrix; over $\mathbb{R}$, $\mathbf{L}^\mathsf{H} = \mathbf{L}^\mathsf{T}$ |

## 6 Differential Equations

| Notation | Meaning |
|---|---|
| $y'$ | First derivative |

| Notation | Meaning |
|---|---|
| $y''$ | Second derivative |
| $\dot{x}$ | First derivative of $x$ with respect to time |
| $\ddot{x}$ | Second derivative of $x$ with respect to time |
| $F(t, y, y', \ldots, y^{(n)}) = 0$ | General $n$th-order ordinary differential equation |
| $y^{(n)} = f(t, y, \ldots, y^{(n-1)})$ | $n$th-order ordinary differential equation in normal form |
| $y(t_0) = y_0$ | Initial condition for a scalar differential equation |
| $y' + p(t)y = g(t)$ | Standard form of a first-order linear differential equation |
| $\mu(t)$ | Integrating factor; for $y' + p(t)y = g(t)$, one may take $\mu(t) = \exp\left(\int p(t)\ \mathrm{d}t\right)$ up to a nonzero constant factor |
| $M(x, y)\ \mathrm{d}x + N(x, y)\ \mathrm{d}y = 0$ | Differential form of a first-order differential equation |
| $M_x, M_y, N_x, N_y$ | Subscript notation for the corresponding partial derivatives |
| $L[y]$ | Linear differential operator applied to $y$ |
| $L[y] = \sum_{k=0}^{n} a_k(t) y^{(k)}$ | General $n$th-order linear differential operator, with $a_n \not\equiv 0$ |
| $L[y] = 0,\ L[y] = g$ | Homogeneous and nonhomogeneous linear differential equations, respectively |
| $y_h,\ y_c$ | Homogeneous or complementary part of a solution |
| $y_p$ | Particular solution |
| $c_1, c_2, \ldots$ | Arbitrary constants in a general solution |
| $\sum_{k=0}^{n} a_k r^k = 0$ | Characteristic equation for a constant-coefficient homogeneous linear differential equation |
| $r_1, \ldots, r_n$ | Characteristic roots |
| $W(y_1, \ldots, y_n)(t)$ | Wronskian of $y_1, \ldots, y_n$ |
| $y = \sum_{n=0}^{\infty} a_n (x - x_0)^n$ | Power-series form about the ordinary point $x_0$ |
| $y = \sum_{n=0}^{\infty} a_n (x - x_0)^{n+r}$ | Frobenius-series form about a regular singular point $x_0$; $r$ denotes an indicial exponent |
| $\mathbf{x}' = \mathbf{f}(t, \mathbf{x})$ | General first-order system |
| $\mathbf{x}(t_0) = \mathbf{x}_0$ | Initial condition for a first-order system |
| $\mathbf{x}' = \mathbf{A}\mathbf{x} + \mathbf{g}(t)$, $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{g}(t)$ | Linear first-order system in state-vector form |
| $\mathbf{x}' = \mathbf{f}(\mathbf{x})$ | Autonomous first-order system |
| $\mathbf{x}_e$ | Equilibrium point; $\mathbf{f}(\mathbf{x}_e) = \mathbf{0}$ |
| $\mathbf{\Phi}(t)$ | Fundamental matrix of a linear system |
| $\mathcal{L}\{f(t)\}(s),\ \mathcal{L}[f](s)$ | Laplace transform of $f$ |
| $\mathcal{L}^{-1}\{F(s)\}(t),\ \mathcal{L}^{-1}[F](t)$ | Inverse Laplace transform of $F$ |
| $(f * g)(t)$ | One-sided convolution, $(f * g)(t) \coloneqq \int_0^t f(\tau) g(t - \tau)\ \mathrm{d}\tau$ |
| $H(t - a),\ u(t - a)$ | Heaviside or unit-step function shifted to $t = a$; its value at the jump is specified when relevant |
| $\delta(t),\ \delta(t - a)$ | Dirac delta distribution, or unit impulse, centered at 0 or shifted to $t = a$ |
| $t_n = t_0 + nh$ | Uniform numerical grid with step size $h > 0$ |
| $y_n \approx y(t_n)$ | Numerical approximation to the exact solution at $t_n$ |
| $y_{n+1} = y_n + hf(t_n, y_n)$ | Forward (explicit) Euler method for the initial-value problem $y\prime = f(t, y)$ on the grid $t_n = t_0 + nh$ |
| $u_x, u_t, u_{xx}, u_{xy}, u_{tt}, \ldots$ | Subscript notation for partial derivatives of a scalar function $u$ |

| Notation | Meaning |
|---|---|
| $\partial_{\mathbf{n}} u$ | Normal derivative, $\partial_{\mathbf{n}} u \coloneqq \nabla u \cdot \mathbf{n}$, for a unit normal vector $\mathbf{n}$ |
| $u\vert_{\partial\Omega} = g$ | Dirichlet boundary condition on $\partial\Omega$ |
| $\partial_{\mathbf{n}} u\vert_{\partial\Omega} = g$ | Neumann boundary condition on $\partial\Omega$ |
| $\alpha u + \beta\, \partial_{\mathbf{n}} u = g$ | Robin boundary condition on $\partial\Omega$ |
| $a_n,\ b_n$ | Fourier coefficients for the real $2L$-periodic convention $f(x) \sim \frac{a_0}{2} + \sum_{n=1}^{\infty} \left( a_n \cos \frac{n\pi x}{L} + b_n \sin \frac{n\pi x}{L} \right)$ |
| $a_n$ | $a_n = \frac{1}{L} \int_{-L}^{L} f(x) \cos \frac{n\pi x}{L}\, \mathrm{d}x$ for $n \geq 0$ |
| $b_n$ | $b_n = \frac{1}{L} \int_{-L}^{L} f(x) \sin \frac{n\pi x}{L}\, \mathrm{d}x$ for $n \geq 1$ |
| $\widehat{f}(\boldsymbol{\xi}),\ \mathcal{F}f(\boldsymbol{\xi})$ | Fourier transform under the normalization $\widehat{f}(\boldsymbol{\xi}) \coloneqq \int_{\mathbb{R}^n} f(\mathbf{x}) e^{-2\pi i \mathbf{x}\cdot\boldsymbol{\xi}}\, \mathrm{d}\mathbf{x}$; inversion uses $f(\mathbf{x}) = \int_{\mathbb{R}^n} \widehat{f}(\boldsymbol{\xi}) e^{2\pi i \mathbf{x}\cdot\boldsymbol{\xi}}\, \mathrm{d}\boldsymbol{\xi}$ whenever applicable |
| $\mathcal{F}^{-1}$ | Inverse Fourier transform under the stated normalization |
| $-(py')' + qy = \lambda w y$ | Sturm–Liouville eigenvalue equation, where $p$, $q$, and $w$ are functions of $x$ |
| $\lambda_n,\ \phi_n$ | Sturm–Liouville eigenvalue and corresponding eigenfunction |
| $\langle f, g \rangle_w$ | Weighted inner product, $\langle f, g \rangle_w \coloneqq \int_a^b \overline{f(x)} g(x) w(x)\, \mathrm{d}x$ |
| $\langle \phi_m, \phi_n \rangle_w = \delta_{mn}$ | Weighted orthonormality of eigenfunctions |
| $G(x, \xi),\ G(\mathbf{x}, \boldsymbol{\xi})$ | Green's function |
| $\mathrm{Ai}(x),\ \mathrm{Bi}(x)$ | Airy functions |
| $J_\nu(x)$ | Bessel function of the first kind of order $\nu$ |
| $Y_\nu(x),\ N_\nu(x)$ | Bessel function of the second kind of order $\nu$ |
| $I_\nu(x),\ K_\nu(x)$ | Modified Bessel functions of the first and second kinds |
| $H_\nu^{(1)}(x),\ H_\nu^{(2)}(x)$ | Hankel functions of the first and second kinds |
| $P_n(x)$ | Legendre polynomial of degree $n$ |
| $P_\nu(x),\ Q_\nu(x)$ | Legendre functions of the first and second kinds |
| $P_\ell^m(x)$ | Associated Legendre function for integer degree $\ell$ and order $m$ |
| $H_n(x),\ \mathrm{He}_n(x)$ | Physicists' and probabilists' Hermite polynomials, respectively |
| $L_n^{(\alpha)}(x),\ L_n(x)$ | Generalized and ordinary Laguerre polynomials; $L_n(x) = L_n^{(0)}(x)$ |
| $T_n(x),\ U_n(x)$ | Chebyshev polynomials of the first and second kinds |
| $P_n^{(\alpha,\beta)}(x)$ | Jacobi polynomial |
| $C_n^{(\lambda)}(x)$ | Gegenbauer (ultraspherical) polynomial |
| $\mathrm{erf}(x),\ \mathrm{erfc}(x)$ | Error function and complementary error function |
| ${}_2F_1(a, b; c; z)$ | Gauss hypergeometric function |

## 7 Combinatorics and Discrete Mathematics

| Notation | Meaning |
|---|---|
| $[n] = \{1, 2, \ldots, n\}$ | Standard $n$-element set; $[0] = \varnothing$ when needed |
| $n!$ | Factorial |
| $n!!$ | Double factorial: product of positive integers of the same parity not exceeding $n$, with $0!! = (-1)!! = 1$ |

| Notation | Meaning |
|---|---|
| $n^{\underline{k}}$ | Falling factorial, $n(n-1)\cdots(n-k+1)$ |
| $n^{\overline{k}}$ | Rising factorial, $n(n+1)\cdots(n+k-1)$ |
| $(a)_k$ | Rising Pochhammer symbol $a(a+1)\cdots(a+k-1)$ in special-functions contexts |
| ${}_nP_r$ | Number of ordered selections of $r$ distinct elements from an $n$-element set, ${}_nP_r = \dfrac{n!}{(n-r)!}$ for $0 \le r \le n$ |
| $\binom{n}{r}$, ${}_nC_r$ | Number of $r$-element subsets of an $n$-element set |
| $\left(\binom{n}{r}\right)$, ${}_nH_r$ | $r$-combinations with repetition from $n$ types; equals $\binom{n+r-1}{r}$ |
| $\binom{n}{n_1,\ldots,n_k}$ | Multinomial coefficient, for nonnegative integers $n_1,\ldots,n_k$ with $n_1+\cdots+n_k=n$ |
| $\left[\begin{matrix} n \\ k \end{matrix}\right]$, $c(n,k)$ | Unsigned Stirling number of the first kind; permutations of $n$ elements with $k$ cycles |
| $s(n,k)$ | Signed Stirling number of the first kind; $s(n,k) = (-1)^{n-k}c(n,k)$ |
| $\left\{\begin{matrix} n \\ k \end{matrix}\right\}$, $S(n,k)$ | Stirling number of the second kind; partitions of an $n$-element set into $k$ nonempty blocks |
| $D_n$, $!n$ | Number of derangements of $n$ elements; permutations with no fixed points |
| $C_n$ | $n$th Catalan number, $C_n = \dfrac{1}{n+1}\binom{2n}{n}$ for $n \ge 0$ |
| $B_n$ | $n$th Bell number; number of partitions of an $n$-element set |
| $p(n)$ | Number of integer partitions of $n$, with $p(0)=1$ |
| $H_n = \sum_{k=1}^{n} \dfrac{1}{k}$ | $n$th harmonic number for $n \ge 1$, with $H_0 = 0$ |
| $(X, \preceq)$ | Partially ordered set (poset) |
| $\prec$, $\succeq$, $\succ$ | Strict and reverse order relations associated with $\preceq$ |
| $\le_{\text{lex}}$ | Lexicographic order |
| $x \lessdot y$ | Cover relation: $y$ covers $x$ |
| $B_n$ | Boolean lattice (Boolean poset) on an $n$-element ground set |
| $G = (V, E)$ | Graph with vertex set $V$ and edge set $E$ |
| $V(G)$ | Vertex set of $G$ |
| $E(G)$ | Edge set of $G$ |
| $N(v)$, $\Gamma(v)$ | Neighborhood of vertex $v$ |
| $N[v] = N(v) \cup \{v\}$ | Closed neighborhood of the vertex $v$ |
| $\deg_G(v)$, $\deg(v)$ | Degree of vertex $v$ |
| $\deg^+(v)$, $\deg^-(v)$ | Out-degree and in-degree of $v$ in a directed graph |
| $\delta(G)$, $\Delta(G)$ | Minimum and maximum vertex degrees, respectively, of a nonempty finite graph $G$ |
| $d_G(u,v)$, $d(u,v)$ | Graph distance between vertices $u$ and $v$ |
| $G[S]$ | Subgraph of $G$ induced by the vertex set $S \subseteq V(G)$ |
| $G - v$, $G - S$ | Graph obtained by deleting the vertex $v$, or all vertices in $S$, together with their incident edges |
| $G - e$ | Graph obtained by deleting the edge $e$ |
| $G/e$ | Graph obtained by contracting the edge $e$ |
| $\overline{G}$ | Complement of the graph $G$ |
| $K_n$ | Complete graph on $n$ vertices |

| Notation | Meaning |
|---|---|
| $K_{m,n}$ | Complete bipartite graph with parts of sizes $m$ and $n$ |
| $P_n$ | Path graph on $n$ vertices |
| $C_n$ | Cycle graph on $n$ vertices |
| $\mathbf{A}(G),\ \mathbf{A}_G$ | Adjacency matrix of $G$ |
| $\mathbf{D}(G),\ \mathbf{D}_G$ | Degree matrix of $G$, the diagonal matrix of vertex degrees in the chosen vertex order |
| $\mathbf{L}(G),\ \mathbf{L}_G$ | Graph Laplacian, usually $\mathbf{D}(G) - \mathbf{A}(G)$ |
| $\chi(G)$ | Chromatic number of $G$ |
| $P_G(k)$ | Chromatic polynomial of a finite graph $G$; for each nonnegative integer $k$, $P_G(k)$ counts proper vertex colorings with $k$ labeled colors |
| $\nu(G)$ | Matching number of $G$: the maximum size of a matching (a set of pairwise vertex-disjoint edges) |
| $\alpha(G)$ | Independence number of $G$: maximum size of an independent vertex set |
| $\omega(G)$ | Clique number of $G$: maximum size of a complete vertex set |
| $\mathrm{Aut}(G)$ | Automorphism group of the graph $G$ |
| $\tau(G),\ T(G)$ | Number of spanning trees of $G$ |
| $R(k,\ell),\ r(k,\ell),\ R(k),\ r(k)$ | Ramsey numbers; $R(k) = R(k,k)$ in the diagonal case |
| $A(x) = \sum_{n=0}^{\infty} a_n x^n$ | Ordinary generating function of the sequence $(a_n)_{n\geq 0}$ |
| $\sum_{n=0}^{\infty} a_n \frac{x^n}{n!}$ | Exponential generating function of the sequence $(a_n)_{n\geq 0}$ |
| $[x^n]F(x)$ | Coefficient of $x^n$ in the formal power series or generating function $F(x)$ |

## 8 Probability and Statistics

Parameter symbols are local to the relevant context: for gamma distributions, $\beta$ denotes a rate parameter and $\theta$ a scale parameter, so $\theta = 1/\beta$, while in hypothesis testing $\beta$ denotes a Type II error probability. The geometric distribution counts the number of trials up to and including the first success.

| Notation | Meaning |
|---|---|
| $(\Omega, \mathcal{F}, \mathbb{P})$ | Probability space, with sample space $\Omega$, event $\sigma$-algebra $\mathcal{F}$, and probability measure $\mathbb{P}$ |
| $\omega \in \Omega$ | Outcome or sample point |
| $A \in \mathcal{F}$ | Event |
| $X \colon \Omega \to \mathbb{R}$ | Real-valued random variable; measurability from $(\Omega, \mathcal{F})$ to $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ is understood |
| $\mathbb{P}(A)$ | Probability of event $A$ |
| $\mathbb{P}(A \mid B)$ | Conditional probability of $A$ given $B$ |
| $X \perp\!\!\!\perp Y$ | $X$ and $Y$ are independent. |
| $X \perp\!\!\!\perp Y \mid Z$ | $X$ and $Y$ are conditionally independent given $Z$. |
| $p_X(x) = \mathbb{P}(X = x)$ | Probability mass function (pmf) of a discrete random variable $X$ |
| $f_X(x)$ | Probability density function (pdf) of $X$ |
| $F_X(x) = \mathbb{P}(X \leq x)$ | Cumulative distribution function (cdf) of $X$ |
| $p_{X,Y}(x,y),\ f_{X,Y}(x,y)$ | Joint pmf or joint pdf of $(X, Y)$ |
| $p_{X\mid Y}(x \mid y),\ f_{X\mid Y}(x \mid y)$ | Conditional pmf or conditional pdf of $X$ given $Y = y$ |

| Notation | Meaning |
|---|---|
| $F^{-1}(p)$ | Generalized inverse of a cdf $F$, $F^{-1}(p) = \inf\{x \in \mathbb{R} \mid F(x) \geq p\}$; the $p$-quantile, for $0 < p < 1$ |
| $\mathbb{E}[X]$ | Expectation of random variable $X$ |
| $\mathbb{E}[X \mid Y]$ | Conditional expectation of $X$ given $Y$ |
| $\mathbb{E}[X \mid \mathcal{G}]$ | Conditional expectation of $X$ with respect to a sub-$\sigma$-algebra $\mathcal{G} \subseteq \mathcal{F}$ |
| $\mathrm{Var}(X)$ | Variance of $X$ |
| $\sigma_X = \sqrt{\mathrm{Var}(X)}$ | Population standard deviation of $X$ |
| $\mathrm{Cov}(X, Y)$ | Covariance of $X$ and $Y$ |
| $\mathbf{\Sigma} = \mathrm{Cov}(\mathbf{X})$ | Covariance matrix of a random vector $\mathbf{X} = (X_1, \ldots, X_n)$ with finite second moments; its $(i, j)$-entry is $\mathrm{Cov}(X_i, X_j)$ |
| $\rho_{X,Y},\ \mathrm{Corr}(X, Y)$ | Pearson correlation coefficient of $X$ and $Y$ when both have finite positive variances |
| $r$ | Sample correlation coefficient |
| $\varphi_X(t) = \mathbb{E}[e^{itX}]$ | Characteristic function of $X$ |
| $M_X(t) = \mathbb{E}[e^{tX}]$ | Moment-generating function (mgf) of $X$, when finite in a neighborhood of 0 |
| $\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$ | Sample mean as a statistic of a random sample $X_1, \ldots, X_n$ |
| $\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$ | Observed sample mean for data $x_1, \ldots, x_n$ |
| $S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2$ | Sample variance as a statistic, for $n \geq 2$ |
| $s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2$ | Observed sample variance, for $n \geq 2$ |
| $S = \sqrt{S^2},\ s = \sqrt{s^2}$ | Sample standard deviation as a statistic and its observed value |
| $\hat{\theta}$ | Estimator or estimate of a parameter $\theta$ |
| $\mathrm{Bias}(\hat{\theta})$ | Bias of a real-valued estimator of $\theta$: $\mathbb{E}[\hat{\theta}] - \theta$, when the expectation exists |
| $\mathrm{MSE}(\hat{\theta})$ | Mean squared error of a real-valued estimator of $\theta$: $\mathbb{E}[(\hat{\theta} - \theta)^2]$, when finite |
| $\hat{p}$ | Estimator or estimate of a probability parameter $p$; commonly the sample proportion |
| $\mathrm{SE}(\hat{\theta})$ | Standard error of an estimator $\hat{\theta}$ |
| $L(\theta; \mathbf{x})$ | Likelihood function for parameter $\theta$ given observed data $\mathbf{x}$ |
| $\ell(\theta; \mathbf{x}) = \ln L(\theta; \mathbf{x})$ | Log-likelihood function |
| $H_0$ | Null hypothesis |
| $H_1,\ H_a$ | Alternative hypothesis |
| $\alpha$ | Significance level; an upper bound on the Type I error probability under the specified test |
| $\beta$ | Type II error probability under a specified alternative |
| $\nu$ | Degrees of freedom |
| $X \sim \mathrm{Bern}(p)$ | Bernoulli distribution with success probability $p$ |
| $X \sim \mathrm{Bin}(n, p)$ | Binomial distribution |
| $X \sim \mathrm{Hypergeom}(N, K, n)$ | Hypergeometric distribution: number of successes in $n$ draws without replacement from $N$ items, of which $K$ are successes, where $0 \leq K, n \leq N$. For $\max(0, n - N + K) \leq k \leq \min(K, n)$, $\mathbb{P}(X = k) = \frac{\binom{K}{k}\binom{N-K}{n-k}}{\binom{N}{n}}$. |

| Notation | Meaning |
|---|---|
| $X \sim \mathrm{Geom}(p)$ | Geometric distribution on $\mathbb{N}$, counting trials up to and including the first success; $\mathbb{P}(X = k) = (1-p)^{k-1}p$ for $k \in \mathbb{N}$ |
| $X \sim \mathrm{Pois}(\lambda)$ | Poisson distribution with parameter $\lambda$ |
| $X \sim \mathrm{Unif}(a, b)$ | Continuous uniform distribution on $[a, b]$ |
| $X \sim \mathrm{Exp}(\lambda)$ | Exponential distribution with rate $\lambda$ |
| $X \sim \Gamma(\alpha, \beta)$ | Gamma distribution with shape $\alpha > 0$ and rate $\beta > 0$; $f_X(x) = \dfrac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}$ for $x > 0$ |
| $X \sim \Gamma(\alpha, \theta)$ | Gamma distribution with shape $\alpha > 0$ and scale $\theta > 0$; $f_X(x) = \dfrac{1}{\Gamma(\alpha)\theta^\alpha} x^{\alpha-1} e^{-x/\theta}$ for $x > 0$, with $\theta = 1/\beta$ |
| $X \sim \mathrm{Beta}(\alpha, \beta)$ | Beta distribution with shape parameters $\alpha, \beta > 0$ and density $f_X(x) = \dfrac{x^{\alpha-1}(1-x)^{\beta-1}}{B(\alpha, \beta)}$ for $0 < x < 1$ |
| $X \sim \mathcal{N}(\mu, \sigma^2)$, $X \sim N(\mu, \sigma^2)$ | Normal distribution with mean $\mu$ and variance $\sigma^2$ |
| $\phi(x)$ | Standard normal probability density function, $\phi(x) = \dfrac{1}{\sqrt{2\pi}} e^{-x^2/2}$ |
| $\Phi(x)$ | Standard normal cumulative distribution function, $\Phi(x) = \displaystyle\int_{-\infty}^{x} \phi(t)\, \mathrm{d}t$ |
| $X \sim \chi^2_\nu$ | Chi-squared distribution with $\nu$ degrees of freedom |
| $X \sim t_\nu$ | Student's $t$ distribution with $\nu$ degrees of freedom |
| $X \sim F_{\nu_1, \nu_2}$ | $F$ distribution with numerator and denominator degrees of freedom $\nu_1, \nu_2$ |
| $X_n \xrightarrow{\text{a.s.}} X$ | Almost-sure convergence |
| $X_n \xrightarrow{\mathbb{P}} X$ | Convergence in probability |
| $X_n \xrightarrow{L^p} X$ | Convergence in $L^p$ |
| $X \overset{\text{d}}{=} Y$ | Equality in distribution of the random variables $X$ and $Y$ |
| $X_n \xrightarrow{\text{d}} X,\ X_n \Rightarrow X$ | Convergence in distribution |
| $\mathcal{L}(X)$ | Distribution (law) of the random variable $X$ |

## 9 Geometry

In triangle notation, $A, B, C$ denote vertices and $a, b, c$ the opposite side lengths. Lowercase letters such as $\ell, m, n$ may denote lines. The notation $\overrightarrow{AB}$ may denote either the ray from $A$ through $B$ or the vector from $A$ to $B$, according to context. Greek letters such as $\omega, \Omega, \Gamma$ may be used as local names for circles. In differential geometry, the Frenet frame follows the standard uppercase convention $\mathbf{T}, \mathbf{N}, \mathbf{B}$, typeset in bold to emphasize that these are vectors; uppercase italic letters such as $X, Y$ may denote vector fields in the usual field-specific convention.

| Notation | Meaning |
|---|---|
| $\triangle ABC$ | Triangle with vertices $A, B, C$ |
| $\angle ABC$ | Geometric angle with vertex $B$ |
| $m\angle ABC$ | Measure of the angle $\angle ABC$ |
| $\angle(\ell, m)$ | Directed angle from the line $\ell$ to the line $m$, taken modulo $\pi$ |
| $\overline{AB}$ | Segment joining $A$ and $B$ |
| $AB$ | Length of $\overline{AB}$ |
| $\overrightarrow{AB}$ | Ray from $A$ through $B$, or vector from $A$ to $B$, according to context |

| Notation | Meaning |
|---|---|
| $\overrightarrow{AB} = \mathbf{b} - \mathbf{a}$ | Vector from $A$ to $B$ when $\mathbf{a}$ and $\mathbf{b}$ are the corresponding position vectors |
| $\overleftrightarrow{AB}$ | Line through $A$ and $B$ |
| $\ell \parallel m$ | Lines $\ell$ and $m$ are parallel |
| $\ell \perp m$ | Lines $\ell$ and $m$ are perpendicular |
| $\triangle ABC \cong \triangle DEF$ | Congruent triangles; corresponding vertices are listed in the same order |
| $\triangle ABC \sim \triangle DEF$ | Similar triangles; corresponding vertices are listed in the same order |
| $[ABC]$ | Area of $\triangle ABC$ |
| $C(O, r)$ | Circle with center $O$ and radius $r$ |
| $\odot O$ | Circle centered at $O$ |
| $\mathrm{Pow}(P, \omega)$ | Power of $P$ with respect to a circle $\omega$ with center $O$ and radius $r$: $OP^2 - r^2$ |
| $(ABC)$ | Circle through the noncollinear points $A, B, C$; in particular, the circumcircle of $\triangle ABC$ |
| $\widehat{AB}$, $\widehat{ACB}$ | Arcs specified by their endpoints, with an additional point used when needed to distinguish the intended arc |
| $O, I, H, G$ | Circumcenter, incenter, orthocenter, and centroid of a triangle, respectively |
| $R, r$ | Circumradius and inradius of a triangle, respectively |
| $s = \dfrac{a+b+c}{2}$ | Semiperimeter of a triangle with side lengths $a, b, c$ |
| $d(P, Q)$ | Distance between the points $P$ and $Q$ |
| $\mathrm{dist}(P, A)$ | Distance from the point $P$ to the set $A$ |
| $\mathrm{length}(C)$, $\ell(C)$ | Length of the curve $C$ |
| $\mathrm{area}(S)$ | Area of the surface $S$ |
| $\mathrm{vol}(\Omega)$ | Volume of the region $\Omega$ |
| $\mathrm{diam}(A)$ | Diameter of the set $A$ |
| $\mathrm{conv}(A)$ | Convex hull of the set $A$ |
| $\mathrm{aff}(A)$ | Affine hull of the set $A$ |
| $\mathbf{x} = \mathbf{p} + t\mathbf{v}$ | Parametric equation of a line through $\mathbf{p}$ in the direction $\mathbf{v}$ |
| $\mathbf{n} \cdot (\mathbf{x} - \mathbf{p}) = 0$ | Hyperplane through $\mathbf{p}$ with normal vector $\mathbf{n}$ |
| $\mathbb{R}^n$ | $n$-dimensional real Euclidean space |
| $S^n = \{\mathbf{x} \in \mathbb{R}^{n+1} \mid \lVert\mathbf{x}\rVert = 1\}$ | Unit $n$-sphere |
| $S_r^{n-1}(\mathbf{a})$ | Sphere of radius $r$ centered at $\mathbf{a} \in \mathbb{R}^n$ |
| $\mathrm{Isom}(X)$ | Isometry group of the metric or geometric space $X$ |
| $\mathbb{RP}^n$ | Real projective $n$-space |
| $\mathbb{CP}^n$ | Complex projective $n$-space |
| $[x_0 : \cdots : x_n]$ | Homogeneous coordinates in projective space |
| $\mathbb{H}^n$ | $n$-dimensional hyperbolic space |
| $d_{\mathbb{H}}$ | Hyperbolic distance |
| $\gamma \colon I \to \mathbb{R}^n$ | Parametrized curve, where $I \subseteq \mathbb{R}$ is an interval |
| $\dot{\gamma}(t)$ | Velocity or tangent vector of a differentiable parametrized curve; for a curve in a manifold $M$, $\dot{\gamma}(t) \in T_{\gamma(t)}M$ |
| $s$ | Arc-length parameter |
| $\mathbf{T}$, $\mathbf{N}$, $\mathbf{B}$ | Unit tangent, principal normal, and binormal vectors of the Frenet frame |
| $\kappa$ | Curvature of a regular curve |
| $\tau$ | Torsion of a space curve |

| Notation | Meaning |
|---|---|
| $T_pM$ | Tangent space of the manifold or regular surface $M$ at $p$ |
| $\mathbf{n}\colon M \to S^2$ | Gauss map of an oriented surface $M \subseteq \mathbb{R}^3$, assigning its unit normal vector |
| I, II | First and second fundamental forms of a regular surface |
| $\kappa_1,\ \kappa_2$ | Principal curvatures of a regular surface |
| $K = \kappa_1\kappa_2$ | Gaussian curvature of a regular surface |
| $H = \dfrac{\kappa_1 + \kappa_2}{2}$ | Mean curvature of a regular surface under the average convention |
| $(M, g)$ | Riemannian manifold $M$ with Riemannian metric $g$ |
| $g_p$ | Riemannian inner product on $T_pM$ |
| $d_g(p, q)$ | Riemannian distance between $p$ and $q$ in the same connected component of $(M, g)$: the infimum of lengths of piecewise $C^1$ curves joining them |
| $\nabla_X Y$ | Covariant derivative of the vector field $Y$ in the direction of the vector field $X$; in Riemannian geometry, $\nabla$ commonly denotes the Levi–Civita connection |
| $\Gamma^k_{ij}$ | Christoffel symbols of a connection in local coordinates |
| $\exp_p\colon \mathcal{D}_p \to M$ | Riemannian exponential map at $p$, where $\mathcal{D}_p \subseteq T_pM$ is its maximal domain. If $M$ is geodesically complete, then $\mathcal{D}_p = T_pM$ |
| $\nabla_{\dot\gamma}\dot\gamma = 0$ | Geodesic equation for an affinely parametrized curve $\gamma$ |

## 10 Number Theory

| Notation | Meaning |
|---|---|
| $a \mid b$ | $a$ divides $b$. |
| $a \nmid b$ | $a$ does not divide $b$. |
| $\gcd(a_1, \ldots, a_n),\ (a, b)$ | Greatest common divisor; $(a, b)$ for two integers when the context is clear |
| $\operatorname{lcm}(a_1, \ldots, a_n)$ | Least common multiple |
| $p^k \parallel n$ | $p^k$ exactly divides $n$: $p^k \mid n$ and $p^{k+1} \nmid n$. |
| $v_p(n),\ \nu_p(n)$ | $p$-adic valuation of $n$ |
| $a \equiv b \pmod n$ | $a$ is congruent to $b$ modulo $n$. |
| $[a]_n,\ \overline{a}$ | Residue class of $a$ modulo $n$; $\overline{a}$ when the modulus is clear |
| $a \bmod n$ | Least nonnegative remainder of $a$ modulo $n$, for $n > 0$ |
| $\mathbb{Z}/n\mathbb{Z}$ | Ring of residue classes modulo $n$ |
| $(\mathbb{Z}/n\mathbb{Z})^\times$ | Multiplicative group of units modulo $n$ |
| $a^{-1} \pmod n$ | Multiplicative inverse of $a$ modulo $n$, for $\gcd(a, n) = 1$ |
| $\operatorname{ord}_n(a)$ | Multiplicative order of $a$ modulo $n$, for $\gcd(a, n) = 1$ |
| $\mathbb{F}_q,\ \mathrm{GF}(q)$ | Finite field with $q$ elements, where $q$ is a prime power |
| $\left(\dfrac{a}{p}\right)$ | Legendre symbol, for an odd prime $p$ |
| $\left(\dfrac{a}{n}\right)$ | Jacobi symbol, for an odd positive integer $n$ |
| $\chi \pmod q$ | Dirichlet character modulo $q$ |
| $\varphi(n)$ | Euler totient function |
| $\lambda(n)$ | Carmichael function: exponent of the finite abelian group $(\mathbb{Z}/n\mathbb{Z})^\times$; the least positive $m$ such that $a^m \equiv 1 \pmod n$ for all $a$ coprime to $n$ |
| $B_n$ | $n$th Bernoulli number, under the convention $B_1 = -\frac{1}{2}$ |

| Notation | Meaning |
|---|---|
| $\sum_{d\mid n}$ | Sum over the positive divisors $d$ of $n$ |
| $\prod_{p\mid n}$ | Product over the prime divisors $p$ of $n$ |
| $\sigma_k(n) = \sum_{d\mid n} d^k$ | Sum of the $k$th powers of the positive divisors of $n$ |
| $\tau(n) = \sigma_0(n),\ d(n)$ | Number of positive divisors of $n$ |
| $\sigma(n) = \sigma_1(n)$ | Sum of positive divisors of $n$ |
| $\mu(n)$ | Möbius function |
| $\omega(n)$ | Number of distinct prime divisors of $n$ |
| $\Omega(n)$ | Number of prime factors of $n$, counted with multiplicity |
| $\mathrm{rad}(n)$ | Product of the distinct prime divisors of $n$ |
| $(f * g)(n) = \sum_{d\mid n} f(d)g(n/d)$ | Dirichlet convolution of arithmetic functions $f$ and $g$ |
| $\pi(x)$ | Number of primes not exceeding $x$ |
| $\Lambda(n)$ | Von Mangoldt function: $\ln p$ if $n = p^k$ for a prime $p$ and an integer $k \geq 1$, and 0 otherwise |
| $L(s, \chi)$ | Dirichlet $L$-function, $L(s,\chi) = \sum_{n=1}^{\infty} \frac{\chi(n)}{n^s}$ for $\Re(s) > 1$ |
| $[a_0; a_1, a_2, \dots]$ | Simple continued fraction |
| $[a_0; a_1, \dots, a_n] = \frac{p_n}{q_n}$ | $n$th convergent of a continued fraction |
| $\Phi_n(x)$ | $n$th cyclotomic polynomial |
| $\mathbb{Z}[i]$ | Ring of Gaussian integers |
| $K = \mathbb{Q}(\sqrt{d})$ | Quadratic field, with squarefree $d \in \mathbb{Z}$ and $d \neq 1$ |
| $\mathcal{O}_K$ | Ring of integers of the number field $K$ |
| $\mathrm{N}(\mathfrak{a}) = \lvert\mathcal{O}_K/\mathfrak{a}\rvert$ | Absolute norm of a nonzero integral ideal $\mathfrak{a} \subseteq \mathcal{O}_K$ |
| $D_K,\ \mathrm{disc}(K)$ | Discriminant of the number field $K$ |
| $\mathrm{Cl}(K)$ | Ideal class group of the number field $K$ |
| $h_K = \lvert\mathrm{Cl}(K)\rvert$ | Class number of the number field $K$ |

## 11 Algebra

Unless otherwise stated, rings are associative and have an identity, but need not be commutative. Ring homomorphisms preserve identities, ideals are two-sided, and modules are left modules.

| Notation | Meaning |
|---|---|
| $G$ | Group |
| $e,\ 1_G$ | Identity element of a group |
| $0_G,\ -g$ | Identity and inverse notation for an additively written group $G$ |
| $g^{-1}$ | Inverse of $g \in G$ |
| $\lvert G\rvert$ | Order (cardinality) of $G$ |
| $\mathrm{ord}(g),\ \lvert g\rvert$ | Order of the element $g$ |
| $\langle S\rangle$ | Subgroup generated by $S \subseteq G$ |
| $\langle g\rangle$ | Cyclic subgroup generated by $g$ |
| $H \leq G$ | $H$ is a subgroup of $G$. |
| $H \trianglelefteq G$ | $H$ is a normal subgroup of $G$. |
| $gH,\ Hg$ | Left and right cosets of $H$ in $G$ |
| $[G : H]$ | Index of $H$ in $G$ |

| Notation | Meaning |
|---|---|
| $G/H$ | Set of left cosets of $H \leq G$ in $G$; a quotient group when $H \trianglelefteq G$ |
| $\ker \varphi$ | Kernel of the homomorphism $\varphi$ |
| $\operatorname{im} \varphi$ | Image of the homomorphism $\varphi$ |
| $\operatorname{Hom}(G, H)$ | Set of group homomorphisms from $G$ to $H$ |
| $\operatorname{Aut}(G)$ | Automorphism group of $G$ |
| $\operatorname{Inn}(G)$ | Inner automorphism group of $G$ |
| $\operatorname{Out}(G)$ | Outer automorphism group $\operatorname{Aut}(G)/\operatorname{Inn}(G)$ |
| $Z(G)$ | Center of $G$ |
| $C_G(A),\ C_G(x)$ | Centralizer in $G$ of $A \subseteq G$ or $x \in G$ |
| $N_G(A),\ N_G(H)$ | Normalizer in $G$ of $A \subseteq G$ or $H \leq G$ |
| $g^G = \{hgh^{-1} \mid h \in G\}$ | Conjugacy class of $g \in G$ |
| $[x, y] \coloneqq x^{-1}y^{-1}xy$ | Commutator of $x$ and $y$ |
| $[G, G],\ G'$ | Commutator (derived) subgroup of $G$ |
| $G^{\text{ab}} = G/[G, G]$ | Abelianization of $G$ |
| $\operatorname{Syl}_p(G)$ | Set of Sylow $p$-subgroups of $G$ |
| $n_p$ | Number of Sylow $p$-subgroups of $G$ |
| $G \curvearrowright X$ | Action of $G$ on $X$ |
| $g \cdot x$ | Action of $g \in G$ on $x \in X$ |
| $G \cdot x,\ \operatorname{Orb}_G(x),\ \operatorname{Orb}(x)$ | Orbit of $x$ under the action of $G$ |
| $G\backslash X,\ X/G$ | Set of orbits of a left $G$-action on $X$; $X/G$ is also used when the action convention is clear |
| $\operatorname{Stab}_G(x),\ G_x$ | Stabilizer of $x$ in $G$ |
| $\operatorname{Fix}(g),\ X^g$ | Fixed-point set $\{x \in X \mid g \cdot x = x\}$ of $g \in G$ acting on $X$ |
| $X^H$ | Common fixed-point set $\{x \in X \mid h \cdot x = x \text{ for all } h \in H\}$ of $H \leq G$ |
| $G \times H$ | Direct product of groups |
| $N \rtimes_\varphi H$ | Semidirect product determined by $\varphi\colon H \to \operatorname{Aut}(N)$, with $N$ the normal factor |
| $\langle S \mid R \rangle$ | Group presentation with generators $S$ and relations $R$ |
| $C_n$ | Cyclic group of order $n$ |
| $D_n$ | Dihedral group of symmetries of a regular $n$-gon; $\lvert D_n \rvert = 2n$ (alternative convention: $D_{2n}$) |
| $\operatorname{Sym}(X)$ | Symmetric group of all permutations of the set $X$ |
| $S_n,\ \mathfrak{S}_n$ | Symmetric group on $n$ symbols |
| $(a_1\, a_2\, \ldots\, a_k)$ | Cycle notation for a permutation sending $a_i$ to $a_{i+1}$ for $i < k$ and $a_k$ to $a_1$ |
| $A_n,\ \mathfrak{A}_n$ | Alternating group on $n$ symbols |
| $V_4$ | Klein four-group |
| $Q_8$ | Quaternion group of order 8 |
| $\operatorname{GL}_n(F)$ | General linear group over $F$ |
| $\operatorname{SL}_n(F)$ | Special linear group over $F$ |
| $\operatorname{O}(n)$ | Orthogonal group of $n \times n$ real matrices |
| $\operatorname{SO}(n)$ | Special orthogonal group; matrices in $\operatorname{O}(n)$ with determinant 1 |
| $\operatorname{U}(n)$ | Unitary group of $n \times n$ complex matrices |
| $\operatorname{SU}(n)$ | Special unitary group; matrices in $\operatorname{U}(n)$ with determinant 1 |
| $R$ | Ring |
| $0_R,\ 1_R$ | Additive and multiplicative identities of $R$ |
| $R^\times$ | Group of units of $R$ |

| Notation | Meaning |
|---|---|
| $I,\ J,\ \mathfrak{a},\ \mathfrak{b}$ | Ideals of $R$; unless otherwise stated, ideals are two-sided |
| $\mathfrak{p},\ \mathfrak{q}$ | Prime ideals |
| $\mathrm{Spec}(R)$ | Prime spectrum of a commutative ring $R$: the set of its prime ideals |
| $\mathfrak{m},\ \mathfrak{n}$ | Maximal ideals |
| $\sqrt{I}$ | Radical of the ideal $I$ in a commutative ring $R$: $\{r \in R \mid r^n \in I \text{ for some } n \geq 1\}$ |
| $(a)$ | Two-sided ideal generated by $a$; in a commutative ring, the usual principal ideal |
| $(a_1, \ldots, a_n)$ | Two-sided ideal generated by $a_1, \ldots, a_n$ |
| $I + J$ | Sum of the ideals $I$ and $J$ |
| $IJ$ | Product of the ideals $I$ and $J$ |
| $R/I$ | Quotient ring |
| $R[x],\ R[x_1, \ldots, x_n]$ | Polynomial rings over $R$ in commuting central indeterminates |
| $R[\alpha]$ | Subring generated by $R$ and $\alpha$ in a commutative $R$-algebra |
| $\deg f$ | Degree of the polynomial $f$ |
| $\mathrm{disc}(f)$ | Discriminant of the polynomial $f$ |
| $M_n(R)$ | Ring of $n \times n$ matrices over $R$ |
| $R[G]$ | Group ring of $G$ over $R$ |
| $\mathrm{Frac}(R)$ | Fraction field of the integral domain $R$ |
| $S^{-1}R$ | Localization of a commutative ring $R$ at a multiplicatively closed subset $S \subseteq R$ |
| $R_{\mathfrak{p}}$ | Localization of a commutative ring $R$ at a prime ideal $\mathfrak{p}$: $(R \setminus \mathfrak{p})^{-1}R$ |
| $\mathrm{char}\, R$ | Characteristic of $R$ |
| $\mathbb{H}$ | Quaternion division algebra over $\mathbb{R}$ |
| $M,\ N$ | Left $R$-modules, unless otherwise stated |
| ${}_RM,\ M_R$ | Left and right $R$-modules, when the side is emphasized |
| $M/N$ | Quotient module, for a submodule $N$ of $M$ |
| $M \oplus N$ | Direct sum of modules |
| $\bigoplus_{i \in \Lambda} M_i$ | Direct sum of an indexed family of modules, with index set $\Lambda$ |
| $R^n$ | Free $R$-module of rank $n$ |
| $\mathrm{Hom}_R(M, N)$ | Set of $R$-module homomorphisms from $M$ to $N$ |
| $\mathrm{End}_R(M)$ | Endomorphism ring of the $R$-module $M$ |
| $\mathrm{Aut}_R(M)$ | Automorphism group of the $R$-module $M$ |
| $\mathrm{Ann}_R(M)$ | Annihilator of the $R$-module $M$ |
| $M_R \otimes_R {}_RN,\ M \otimes_R N$ | Tensor product over $R$; the first notation emphasizes that $M$ is a right $R$-module and $N$ a left $R$-module. The side distinction is normally suppressed when $R$ is commutative or the required module structures are otherwise clear |
| $K/F$ | Field extension of $F$ by $K$ |
| $[K : F]$ | Degree of the field extension $K/F$ |
| $F(\alpha),\ F(\alpha_1, \ldots, \alpha_n)$ | Fields generated over $F$ by the indicated elements |
| $F[\alpha]$ | Subring (equivalently, $F$-subalgebra) generated by $F$ and $\alpha$ |
| $m_{\alpha,F}(x)$ | Minimal polynomial of $\alpha$ over $F$ |
| $\overline{F}$ | Algebraic closure of $F$ |
| $\mathrm{Gal}(L/K)$ | Galois group of $L/K$ |
| $L^H$ | Fixed field of $H \leq \mathrm{Gal}(L/K)$ |
| $\mathrm{Tr}_{L/K}(\alpha)$ | Field trace of $\alpha \in L$ from $L$ to $K$ |

| Notation | Meaning |
|---|---|
| $\mathrm{N}_{L/K}(\alpha)$ | Field norm of $\alpha \in L$ from $L$ to $K$ |
| $\mathrm{trdeg}_K L$ | Transcendence degree of $L$ over $K$ |

## 12 Topology

| Notation | Meaning |
|---|---|
| $(X, \mathcal{T})$ | Topological space $X$ with topology $\mathcal{T}$ |
| $\mathcal{T}$ | Topology on the underlying set $X$ |
| $\mathcal{B}$ | Basis for a topology |
| $\mathcal{S}$ | Subbasis for a topology |
| $A^\circ,\ \mathrm{int}\, A$ | Interior of $A$ |
| $\overline{A},\ \mathrm{cl}\, A$ | Closure of $A$ |
| $\partial A$ | Boundary of $A$ |
| $\mathrm{ext}\, A$ | Exterior of $A$ in $X$: $\mathrm{int}(X \setminus A) = X \setminus \overline{A}$ |
| $A'$ | Derived set of $A$: the set of its accumulation points in $X$ |
| $(X, d)$ | Metric space $X$ with metric $d$ |
| $\mathcal{T}_A = \{A \cap U \mid U \in \mathcal{T}\}$ | Subspace topology on $A \subseteq X$ |
| $\pi_i \colon \prod_{j \in J} X_j \to X_i$ | Coordinate projection of a product space, for $i \in J$ |
| $X/\!\sim$ | Quotient space when the quotient set is equipped with the quotient topology |
| $q \colon X \twoheadrightarrow X/\!\sim$ | Canonical quotient map |
| $X/A$ | Quotient space obtained by collapsing the subspace $A \subseteq X$ to a point |
| $C(X, Y)$ | Set of continuous maps from $X$ to $Y$ |
| $X \cong Y$ | Homeomorphic spaces |
| $\mathrm{Homeo}(X)$ | Group of self-homeomorphisms of $X$ |
| $\iota \colon A \hookrightarrow X$ | Inclusion map of a subspace $A \subseteq X$ |
| $f \colon X \hookrightarrow Y$ | Topological embedding when explicitly stated; the hook arrow by itself retains its general meaning of injectivity |
| $(X, x_0)$ | Pointed space with basepoint $x_0$ |
| $I = [0, 1]$ | Unit interval, especially for paths and homotopies |
| $\gamma \colon I \to X$ | Path in $X$ |
| $\gamma(0) = \gamma(1) = x_0$ | Loop based at $x_0$ |
| $H \colon X \times I \to Y$ | Homotopy between maps $X \to Y$ |
| $f \simeq g$ | Homotopic maps $f, g \colon X \to Y$ |
| $X \simeq Y$ | Homotopy equivalent spaces |
| $[\gamma]$ | Homotopy class of a path or loop $\gamma$, according to context |
| $\pi_0(X)$ | Set of path-connected components of $X$; not generally a group |
| $\pi_1(X, x_0)$ | Fundamental group of $X$ based at $x_0$ |
| $\pi_n(X, x_0)$ | $n$th homotopy group of the pointed space $(X, x_0)$, for $n \geq 1$ |
| $f_*$ | Homomorphism induced by a map $f$ on a covariant invariant such as homotopy or homology, with the invariant clear from context |
| $f^*$ | Homomorphism induced by a map $f$ on a contravariant invariant such as cohomology, with the invariant clear from context |
| $p \colon \widetilde{X} \to X$ | Covering map, with $\widetilde{X}$ a covering space of $X$ |

| Notation | Meaning |
|---|---|
| $\widetilde{X}$ | Covering space; it denotes a universal cover only when explicitly specified |
| $\mathrm{Deck}(p)$ | Group of deck transformations of the covering $p$ |
| $D^n = \{\mathbf{x} \in \mathbb{R}^n \mid \|\mathbf{x}\| \leq 1\}$ | Closed $n$-disk |
| $X \vee Y$ | Wedge sum of pointed spaces $X$ and $Y$ |
| $T_0, T_1, T_2, T_3, T_4$ | Separation axioms; here $T_2$ means Hausdorff, $T_3$ means $T_1$ and regular, and $T_4$ means $T_1$ and normal |
| $M^n$ | $n$-dimensional topological manifold |
| $\partial M$ | Boundary of a manifold $M$ |
| $\Sigma_g$ | Closed connected orientable surface of genus $g$ |
| $M \# N$ | Connected sum of manifolds $M$ and $N$ |
| $\chi(X)$ | Euler characteristic of $X$ |
| $H_n(X; G)$ | $n$th homology group of $X$ with coefficients in $G$ |
| $H_n(X, A; G)$ | $n$th relative homology group of a pair $(X, A)$ with coefficients in an abelian group $G$, where $A \subseteq X$ |
| $H_n(X)$ | Integral homology group $H_n(X; \mathbb{Z})$ when coefficients are omitted |
| $\widetilde{H}_n(X; G)$ | $n$th reduced homology group of $X$ with coefficients in $G$ |
| $H^n(X; G)$ | $n$th cohomology group of $X$ with coefficients in $G$ |
| $b_n(X)$ | $n$th Betti number; for example, $b_n(X) = \dim_{\mathbb{Q}} H_n(X; \mathbb{Q})$ when this dimension is finite |

## 13 Miscellaneous Notation

| Notation | Meaning |
|---|---|
| $\lfloor x \rfloor$ | Floor of $x$ |
| $\lceil x \rceil$ | Ceiling of $x$ |
| $\{x\}$ | Fractional part $x - \lfloor x \rfloor$, when unambiguous |
| $\mathrm{sgn}(x)$ | Sign of a real number or permutation, according to context |
| $a \propto b$ | $a$ is proportional to $b$; in a functional context, the proportionality factor is independent of the variable under discussion. |
| $O(g(n))$ | Big-O upper asymptotic bound |
| $o(g(n))$ | Little-o strict asymptotic bound |
| $\Theta(g(n))$ | Two-sided asymptotic order |
| $\Omega(g(n))$ | Asymptotic lower bound |
| $f \lesssim g,\ f \gtrsim g$ | Upper and lower bounds up to a positive constant factor, respectively, in a stated regime; for nonnegative functions, $f \lesssim g$ means $f(x) \leq C g(x)$ for some constant $C > 0$ |
| $f(x) \asymp g(x)$ | Comparable up to positive multiplicative constants in a stated regime: $c\, g(x) \leq f(x) \leq C\, g(x)$ for some $0 < c \leq C$ and nonnegative $g$ |
| $\sum_{\mathrm{cyc}} f(a, b, c)$ | Cyclic sum over the cyclic permutations of the displayed variables; for three variables, $f(a, b, c) + f(b, c, a) + f(c, a, b)$ |
| $\sum_{\mathrm{sym}} f(a, b, c)$ | Symmetric sum over the distinct permutations of the displayed variables in the summand; the variables and any normalization are stated when needed |
| $\Delta q$ | Finite change in a quantity $q$ |
| $\delta_{ij}$ | Kronecker delta: 1 if $i = j$ and 0 otherwise |

| Notation | Meaning |
|---|---|
| $\epsilon_{ijk}$ | Three-dimensional Levi–Civita symbol: the sign of the permutation $(i, j, k)$ when $(i, j, k)$ is a permutation of $(1, 2, 3)$, and 0 otherwise |